On Accelerating Inviscid Vortical Calculations

A Mathematical Perspective on the Fast Multipole and Vortex Lattice Methods

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1 Introduction

This thesis seeks to provide a mathematical foundation for the unsteady step of calculating the induced velocity in the vortex lattice method, and its computational acceleration through the fast multipole method. The treatment of this matter leverages differential geometry as the framework within which this foundation is laid.

1.1 Motivation

Many modern implementations of calculations on wind turbines invoke archaic methods illsuited for the needs of the industry for which this implementation as a service is provided. Such methods include, but are not limited to, the plurality of steady vortex methods attributed to Ludwig PRANDTL. Admittedly, these methods, now a century old, were perfectly fine in those days of yore before the invention of the digital computer. But now, in the age of personal computers able to run massive fluid simulations using the finite volume method and various flavors of eddy models, those methods are wholly unnecessary idiosyncrasies. All the same, the methods of computational fluid dynamics—the finite volume method in particular—are absurdly time consuming and computationally costly. Thus, concessions must be made between more temporally demanding though numerically accurate predictions, and predictions far more efficient yet fallible to provide such service. It is the goal of this thesis the examine exactly where such concessions ought to be made, and to what degree the predictions thereof are to be trusted. For in an effort to get with the times, so to speak, the main supervisor for this thesis has commisioned a survey of sorts of the implementation of one such method that has become increasingly more popular in its application to wind turbine rotor calculations. It is then imperative to not only manage to implement themethod and test its veracity, but to insure the veracity from the outset. This method is the unsteady *vortex lattice method*, and we shall attempt to accelerate it using the *fast multipole method*.

Although the mathematical theory needed to implement modelling of fluid flow with any semblance of rigor and defensibility is wellestablished and widely adopted in communities where it is warranted—like those of the finite element method—the vortex lattice method we shall consider in this work apparently does not necessarily warrant it in the same manner. The observation has been made that most derivations of the frankly beautiful theory of this method is largely treated in a superficial manner.

1.2 Background

The study of vortical fluid flow started with WHOM?

Panel methods with MORINO

With the advent of the fast multipole method in 1987 by Leslie F. GREENGARD and Vladimir V. ROHLIN, the unsteady vortex lattice method could be improved dramatically.

1.3 Overview

This thesis is sectioned off into distinct parts, which in large part built upon one another. Introductionally, the theory of differential geometry is provided in a synoptic manner, from which we shall build a sense of what we mean by *fluids*, and potential and vortical flows thereof. Here we also discuss the solution of these flows in terms of the fluid velocity vector field. As we shall exploit some results of complex analysis, we shall frame that theory under this very same framework, which we shall extend to the theory of twodimensional airfoils and flow thereover. A thorough discussion on the theory of spherical harmonics and its application to multipole expansions of the GREEN function follows, along with the translatory theory thereof. We shall then discuss the boundary element method, and how it simplifies to the vortex lattice method. A proposal to improve the computational performance of this by multipole expasions is explored, followed by its numerical implementation in `Python`.

2 Vector fields

Throughout this text, we shall draw from the results of the theory of differential geometry, the most important of which is *the fundamental theorem of multivariate calculus*. The conditions under which this theorem holds is of such importance to the validity of this work that some of the underlying theory will be discussed. Since a thorough understanding of the theory requires such attention to detail in order to discuss in any more meaningful detail than merely superficial mention, only the minimal required vocabulary and ideas to grasp will be introduced. For a more thorough derivation and explanation of the formalities of the theory and the background thereof, the reader is therefore referred to a dedicated text on the topic. The work at hand will utilize the conventions of Shigeyuki MORITA's *Geometry of Differential Forms* for this purpose, and any rigorous proofs or additional definitions to that end are referred thither. Admittedly, not including proofs of what is stated to be such foundational results preceding this work will not serve to enhance intuition *per se*, and so the reason for its inclusion is in part due to completeness, and in part so as to insure ourselves that there is indeed work from which we can draw already proven rules of arithmetic. Subsequently, the theory of functions of a complex variable will be discussed. For although the work at hand is concerned with fluid flow of three dimensions, results from the study of fluid flows of two dimensions will turn out to be useful, which we shall find can be viewed through the lens of complex analysis.

It ultimately is results pertaining to the fluid flow of air around and in the wake of a wind turbine rotor we wish to model. Of pertinence, then, is to outline the mathematical description of fluid flow altogether. For the purposes of this text, it is sufficient we establish for the moment a fluid to be a volume of a specified density and viscosity, both of which govern its dynamics, or motion. We suppose this motion may be represented in terms of a velocity vector of scalar functions in the infinite Euclidean space it occupies. In practice, such a vector is found through the integration of a differential equation governed by the applicable physical principles. We shall justify this notion of fluids for application to our problem of wind turbines, as it will in time become clear that the assumptions made are in no way general to the physics of fluid flow, or even correct, for that matter.

The aforementioned fundamental theorem of multivariate calculus may then be employed to transform such an integrated differential equation evaluated throughout the entire fluid domain to one over its boundary. In the case of a wind turbine blade, this entails integrating over its surface. There are two special cases of this fundamental theorem of particular interest, namely the GAUSS–OSTROGRADSKY theorem, and the THOMSON–STOKES theorem. The names are inconsequential, as they are indeed only merely offshoots of this more fundamental theorem, and are as such simply often labeled the divergence theorem and curl theorem. Unsurprisingly, they respectively relate the divergence and curl in the fluid to some flux of fluid at the boundary, and are given by

$$\int_{\Omega} \operatorname{div} \boldsymbol{u} \, \mathrm{d}V = \int_{\partial\Omega} \boldsymbol{u} \cdot \mathrm{d}\boldsymbol{S}, \qquad \int_{S} \operatorname{curl} \boldsymbol{u} \cdot \mathrm{d}\boldsymbol{S} = \int_{\partial S} \boldsymbol{u} \cdot \mathrm{d}\boldsymbol{s}. \tag{2.1a, b}$$

As is stated, the derivation of this fundamental theorem is beyond the scope of this work, so it will simply be presented once the foundation has been laid. This foundation will also reveal the theory of harmonic analysis, upon which we may build complex analysis. Once discussed, the topic of fluid mechanics will at last be addressed in full, and we shall discuss its dynamics and kinematics. Finally, we shall derive the theory with which this thesis seeks to address its problems—namely the decomposition of fluid processes in solenoidal and vortical processes.

2.1 Exterior calculus

The majority of the theoretical background in this work will hinge almost in its entirety on the validity of the equations (2.1), and theory sufficient for understanding their applications will be discussed in this section.

> **Definition 2.1** (RIEMANN manifold)**.** An n-dimensional manifold Ω is a topological space locally similar to Euclidean space $\mathbb{E}^n$, in a smooth sense. It is said to be of the RIEMANN kind if it is equipped with the positive definite RIEMANN metric $\mathfrak{g} = \mathrm{g}_{ij}$.

The RIEMANN metric is left purposefully unspecified, as it depends on the representation of the manifold chosen. Throughout this work, we shall work on the oriented RIEMANN manifold Ω of dimension three, Euclidean space in particular, but we shall for the moment consider one of dimension n so as to speak of the details in general. The intrinsic smoothness of the manifold reveals something about its differentiability, which is defined by the tangent space $\mathrm{T}_p\,\Omega$ of some point p on the manifold. We shall then admit the definition that a vector field on Ω to be a smooth section of the tangent bundle $\mathrm{T}\,\Omega$, the disjoint union of the tangent spaces on the manifold. At each point $p \in \Omega$, the value of such a vector field then lies in the tangent space.[1,2,3] We shall make no distinction in notation between the tangent bundle or sections thereof. Defining an inner product on the space of vector fields, $\langle \boldsymbol{a}, \boldsymbol{b} \rangle = \mathrm{g}_{ij} a^i b^j$, where EINSTEIN summation is implied, induces the metric.

It shall often be of great importance to distinguish on what part of this manifold we are working, as the method we shall employ is of the boundary element family. An imbedded manifold is a manifold of lower dimension that is not selfintersective, he boundary of the manifold, in particular, being the most natural such imbedded manifold we shall consider. The following inclusion map provides the precise mechanism by which differential forms in the ambient manifold may be integrated over the imbedded manifold.[4]

Definition 2.2 (Inclusion map)**.** Let $M \subset \Omega$ be an imbedded submanifold. The inclusion is defined as the map

$$\iota_M : M \hookrightarrow \Omega, \qquad \iota_M(p) = p. \tag{2.2}$$

The *pushforward* $(\iota_M)_* : \mathrm{T}_p\, M \to \mathrm{T}_{\iota_M(p)}\,\Omega$ regards the tangent vector in M a tangent vector in Ω. Let $\mathfrak{a} \in \Lambda^k\Omega$, and denote the *pullback* as

$$(\iota_M^* \mathfrak{a})_p(\boldsymbol{u}_1, \dots, \boldsymbol{u}_k) = \mathfrak{a}_{\iota_M(p)}\big((\iota_M)_* \boldsymbol{u}_1, \dots, (\iota_M)_* \boldsymbol{u}_k\big), \qquad \boldsymbol{u}_i \in \mathrm{T}_p\, M, \tag{2.3}$$

restricting $\mathfrak{a}$ to M.

We shall not find the fundamental theorem of multivariate calculus in sections of the tangent bundle. Instead, we endow its dual, the cotangent bundle $\Lambda\Omega \equiv \mathrm{T}^*\,\Omega$, the exterior product $\wedge$. The exterior product may appear obtuse when first presented in the abstract manner we will shortly do, but rest assured it does indeed have a geometric interpretation. For, in an effort to unite the exterior algebra of Hermann GRASSMANN[5] and the quaternion algebra of William Rowan HAMILTON,[6] William Kingdon CLIFFORD laid the foundation on which the theory of the nominal CLIFFORD algebras were built.[7] On such algebras, the exterior product of k covectors is understood to be the oriented volume of the k-parallelotope bound by them,[8] and is just one part of the so-called *geometric product*, which provides a coherent definition of a vector product in general. We shall only draw from this intuition at a later point, as we shall focus on the properties of the exterior product with respect to objects on sections of the cotangent bundle.

The repeated application of the exterior product admits higher grade covector fields. We will therefore use for a grade k monomial section of the exterior algebra $\mathfrak{a} \in \Lambda^k\Omega$ a k-form, and avoid the terminology of covectors altogether, as forms serve a much broader purpose to our end. The exterior product is alternating in the sense that for $\mathfrak{a}_i \in \Lambda\Omega$, given a permutation σ of the positive integers up to k, we have $\mathfrak{a}_{\sigma(1)} \wedge \dots \wedge \mathfrak{a}_{\sigma(k)} = \operatorname{sgn}(\sigma)\left(\mathfrak{a}_1 \wedge \dots \wedge \mathfrak{a}_k\right)$. The sign of a permutation $\operatorname{sgn}(\sigma)$ is 1 if the number of transpositions from the identity permutation is even, and -1 if the number of transpositions is odd. This alternating property further reduces to the two properties that

$$\mathfrak{a} \wedge \mathfrak{a} \equiv 0, \qquad \mathfrak{a} \wedge \mathfrak{b} \equiv -\mathfrak{b} \wedge \mathfrak{a}, \tag{2.4a, b}$$

for any one-forms $\mathfrak{a}, \mathfrak{b}$. In summary, we define the exterior algebra.

Definition 2.3 (Exterior algebra)**.** Let Ω be an n-dimensional RIEMANN manifold. Let furthermore $\mathrm{T}\,\Omega$ be its tangent bundle with basis ∂_i, and $\Lambda\Omega = \mathrm{T}^*\,\Omega$ the cotangent bundle with basis $\mathrm{d}x^i$. Endow it the exterior product $\wedge : \Lambda^k\Omega \times \Lambda^l\Omega \to \Lambda^{k+l}\Omega$ satisfying the conditions of bilinearity and associativity, in addition to the modified alternating condition of equation (2.4b)

$$\mathfrak{a} \wedge \mathfrak{b} = (-1)^{kl}\,\mathfrak{b} \wedge \mathfrak{a}. \tag{2.5}$$

Then the direct sum of bundles $\Lambda^k\Omega$ for $0 \le k \le n$ is the exterior algebra of Ω.

The notion of a tangent space is inherently connected to some sense of differentiability. Thus, given a differentiable curve on the manifold, the tangent vector at a point is described by this differentiability,

[1] [28] FLANDERS, p.54
[2] [46] JOST, p.12
[3] [64] MORITA, pp.169–171
[4] [64] MORITA, pp.34–35,72
[5] [30] GRASSMANN, §55
[6] [38] HAMILTON (1844)
[7] [17] CLIFFORD (1878)
[8] [30] GRASSMANN, pp.50–51

and the partial derivative operators form a basis for the tangent space with respect to local coördinates on the manifold. We may choose for it some local basis $\{\boldsymbol{e}_j\}$, such that a vector field may be expressed $\boldsymbol{u} = u^j \boldsymbol{e}_j$. For $\mathbb{E}^3$, with rectilinear coördinates x, y, z specifically, we shall always write its basis as $\{\partial_x, \partial_y, \partial_z\} \simeq \{\hat{\boldsymbol{\imath}}, \hat{\boldsymbol{\jmath}}, \hat{\boldsymbol{k}}\}$. This *dual space* we discussed earlier may be regarded as the space of all linear functionals from the tangent space onto the field over which the vector field lies. Consider still the general basis vectors $\boldsymbol{e}_j$, along with some dual basis $\{\boldsymbol{e}^i\}$. As a result of the above discussion, we must have that $\boldsymbol{e}^i(\boldsymbol{e}_j) = \delta^i_j$. Clearly,

$$\boldsymbol{e}^i(\boldsymbol{u}) = \boldsymbol{e}^i(u^j \boldsymbol{e}_j) = u^j \boldsymbol{e}^i \boldsymbol{e}_j = u^j \delta^i_j = u^i.$$

In fact, any k-form is by construction a functional over the space of vector fields on the manifold. It is clear that there exists some isomorphism between the vectors and covectors. If $\mathfrak{u} \in \Lambda\Omega$ and $\boldsymbol{u} \in \mathrm{T}\,\Omega$ are isomorphic, these isomorphisms are usually denoted by the sharp $\mathfrak{u}^\sharp = \boldsymbol{u}$ and the flat $\boldsymbol{u}^\flat = \mathfrak{u}$, respectively raising and lowering the indices of the coëfficients.[1] Stated explicitly, with these operators, we may describe any vector field $\boldsymbol{u} = u\hat{\boldsymbol{\imath}} + v\hat{\boldsymbol{\jmath}} + w\hat{\boldsymbol{k}}$ in terms of differential forms as $\boldsymbol{u}^\flat = u\,\mathrm{d}x + v\,\mathrm{d}y + w\,\mathrm{d}z$, so that $u^i \partial_i \simeq u_i\,\mathrm{d}x^i$. This notation however tends to clutter expressions and is cumbersome to read, and it will only be used when strictly necessary. As the astute reader may have noticed, we already readily distinguish between elements of the cotangent and tangent spaces by denoting the former with a fraktur typeface, and the latter with an italic bold. Thus it should be clear from context whichever of the two is meant at any given time.

We note the above expression implies the basis covectors, and indeed reiterate the fact that k-forms themselves are multilinear functions of k vector fields. In fact, when $\mathfrak{a} \in \Lambda\Omega$, we have that $\mathfrak{a}(\boldsymbol{u}) = a_i u^i$, which is a zero-form, or scalar. This generalizes to higher grade forms, with the introduction of the interior product, acting as a contraction in the usual sense of the term.

> **Definition 2.4** (Interior product)**.** Let $\boldsymbol{u} \in \mathrm{T}\,\Omega$, and $\mathfrak{a} \wedge \mathfrak{b} \in \Lambda^{k+l}\Omega$, noting the order of the addition of exponents in the exterior algebra. The interior product $\lrcorner : \mathrm{T}\,\Omega \times \Lambda^{k+l}\Omega \to \Lambda^{k+l-1}$, is given by
>
> $$\boldsymbol{u} \lrcorner (\mathfrak{a} \wedge \mathfrak{b}) = (\boldsymbol{u} \lrcorner \mathfrak{a}) \wedge \mathfrak{b} + (-1)^k \mathfrak{a} \wedge \boldsymbol{u} \lrcorner \mathfrak{b}.$$
>
> For unit $\mathfrak{b}$, we have $\mathfrak{a} \wedge 1 = \mathfrak{a}$, and so we say $\boldsymbol{u} \lrcorner \mathfrak{a} = \mathfrak{a}(\boldsymbol{u})$.

Extrapolating on this idea that the tangent space is inherently dependent on some notion of differentiability forming a basis for the tangent space, the cotangent space must then have a corresponding operation, which we will label the exterior derivative.

> **Definition 2.5** (Exterior derivative)**.** Given a manifold Ω and basis $\{\partial_i\}$, the exterior derivative $\mathrm{d}\colon \Lambda^k\Omega \to \Lambda^{k+1}\Omega$ of a k-form $\mathfrak{a} = a_{\imath}\,\mathrm{d}x^{\imath}$ is given by $\mathrm{d}\mathfrak{a} = \partial_i a_{\imath}\,\mathrm{d}x^i \wedge \mathrm{d}x^{\imath}$, where $\imath$ is a multi-index of i. The product rule is graded, namely $\mathrm{d}(\mathfrak{a} \wedge \mathfrak{b}) = \mathrm{d}\mathfrak{a} \wedge \mathfrak{b} + (-1)^k \mathfrak{a} \wedge \mathrm{d}\mathfrak{b}$. Furthermore, $\mathrm{d}\mathrm{d}\mathfrak{a} \equiv 0$.

In the manipulation of expressions of forms, it shall be necessary to convert forms of a higher grade to one of lower grade, and *vice versa*, for example before or after computing the exterior derivative. To that end, we introduce the HODGE star-operator.

> **Definition 2.6** (HODGE operator)**.** The HODGE operator $\star : \Lambda^k\Omega \to \Lambda^{n-k}\Omega$, where n is the dimension of Ω, such that $\mathfrak{a} \wedge \star\mathfrak{b} = \langle \mathfrak{a}, \mathfrak{b} \rangle\,\mathrm{d}V$. It holds that $\star\star\,\mathfrak{a} = (-1)^{k(n-k)}\mathfrak{a}$.

With the exterior derivative, we may describe the volume of the space with respect to local coördinates. In particular, for a manifold Ω of dimension n, $\mathrm{d}V \equiv \sqrt{\det(\mathfrak{g})}\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n$, where $\det(\mathfrak{g})$ is the determinant of the local metric tensor. We find from the definition of the HODGE star above that $\star\mathrm{d}V = 1$.

In order to study the exterior derivative under the HODGE operator, we define the *codifferential*, given as follows.[2]

> **Definition 2.7** (Coderivative)**.** The codifferential of a k-form is adjoint to the exterior derivative, and is given by
>
> $$\text{đ}\mathfrak{a} = (-1)^k \star^{-1} \mathrm{d} \star \mathfrak{a} = (-1)^{n(k+1)+1} \star \mathrm{d} \star \mathfrak{a}.$$

For illustrative purposes, the relationship between the exterior derivative, coderivative, and HODGE star is given by the following commutative diagram.

[1][66] NISHIKAWA, §3.1

[2][64] MORITA, §4.2

$$
\begin{tikzcd}
\Lambda^k\Omega \arrow[r, "\star"] \arrow[d, "\text{đ}"'] & \Lambda^{n-k}\Omega \arrow[d, "\mathrm{d}"] \\
\Lambda^{k-1}\Omega \arrow[r, "(-1)^k\star"'] & \Lambda^{n-k+1}\Omega
\end{tikzcd}
$$

As we have noted, we are strictly speaking not performing these operations on the space $\Lambda^k\Omega$, but rather sections of it. On these sections we may define an inner product that should be familiar,

$$(\mathfrak{a}, \mathfrak{b}) = \int_\Omega \langle \mathfrak{a}, \mathfrak{b} \rangle \,\mathrm{d}V = \int_\Omega \mathfrak{a} \wedge \star\mathfrak{b}. \tag{2.6}$$

We have of course not defined what the integral of a differential form actually is. Still, assuming inheritance from our familiar notion of the integral as a linear operator, we find that it does indeed constitute an integral analogous to that of the L^2 inner product. This inner product is symmetric, and is the inner product with respect to which the codifferential is adjoint to the exterior derivative, in the sense that $(\mathrm{d}\mathfrak{a}, \mathfrak{b}) = (\mathfrak{a}, \text{đ}\mathfrak{b})$. For proofs and further discussion on the codifferential operator, the reader is again referred to Morita's monograph.[1]

We are now nearing the revelation of the fundamental theorem of multivariate calculus, and with it the end of this section. Before this, however, we ought to finally apply the operators to these differential forms to get a sense of the reason for which they were introduced to begin with. We shall as an example consider the Euclidean vector field $\boldsymbol{u}^\flat = u\,\mathrm{d}x + v\,\mathrm{d}y + w\mathrm{d}z$. Upon calculating its exterior derivative $\mathrm{d}\mathfrak{u}$, one finds that

$$\mathrm{d}\mathfrak{u} = (\partial_x v - \partial_y u)\,\mathrm{d}x \wedge \mathrm{d}y + (\partial_z u - \partial_x w)\,\mathrm{d}z \wedge \mathrm{d}x + (\partial_y w - \partial_z v)\,\mathrm{d}y \wedge \mathrm{d}z.$$

This is nothing less than the two-form corresponding to the curl. Furthermore, the coderivative of a vector field is given by $\text{đ}\mathfrak{u} = -\star\mathrm{d}\star\mathfrak{u}$, which upon calculation one finds to be

$$\text{đ}\mathfrak{u} = -(\partial_x u + \partial_y v + \partial_z w).$$

This is of course the divergence with an opposite sign. And so, we define the operators to be such.

Definition 2.8 (Curl and divergence)**.** Let $\mathfrak{u} \in \Lambda^k\Omega$, with $\dim\Omega = 3$. The curl and divergence of $\mathfrak{u}$ are respectively given by

$$\operatorname{curl}\mathfrak{u} = \star\mathrm{d}\mathfrak{u}, \qquad \operatorname{div}\mathfrak{u} = \star\mathrm{d}\star\mathfrak{u}.$$

From these definitions, it is trivial to prove identities of great utility. For example, $\operatorname{div}\operatorname{curl}\mathfrak{u}$ and $\operatorname{curl}\operatorname{grad}\mathfrak{u}$ both being identically equal to zero. Another way to express the divergence is to consider the contraction of the gradient. We understand the trace of a matrix to be a form of contraction,[2] and the divergence of a vector field is simply the trace of the gradient.[3] For brevity, we take an *ad hoc* approach. Consider the exterior derivative of the interior product $\boldsymbol{u} \lrcorner \,\mathrm{d}V$. We calculate that

$$
\begin{aligned}
\mathrm{d}(\boldsymbol{u} \lrcorner \,\mathrm{d}V) &= \mathrm{d}\sum_i (-1)^{i+1} u_i \mathrm{d}x^1 \wedge \cdots \wedge \widehat{\mathrm{d}x^i} \wedge \cdots \wedge \mathrm{d}x^n \\
&= \partial_i u_i \mathrm{d}V \\
&= \operatorname{div}\boldsymbol{u}\,\mathrm{d}V,
\end{aligned} \tag{2.7}
$$

where $\widehat{\mathrm{d}x^i}$ in the first equality in equation (2.7) means the term is omitted.

Now having the notions of divergence and curl explicitly defined, we are now ready for the progenitor of their named theorems—the fundamental theorem of multivariate calculus. The proof for this theorem is beyond the scope of the present work, so it will only be stated.

Theorem 2.9 (Fundamental Theorem of Multivariate Calculus)**.** *Let* $\mathfrak{u} \in \Lambda^{n-1}\Omega$*, and* $\partial\Omega$ *be the sufficiently regular boundary of* Ω*. Then,*

$$\int_{\partial\Omega} \mathfrak{u} = \int_\Omega \mathrm{d}\mathfrak{u}. \tag{2.8}$$

[1] [64] Morita, pp.153–155

[2] expand on this.

[3] confirm this.

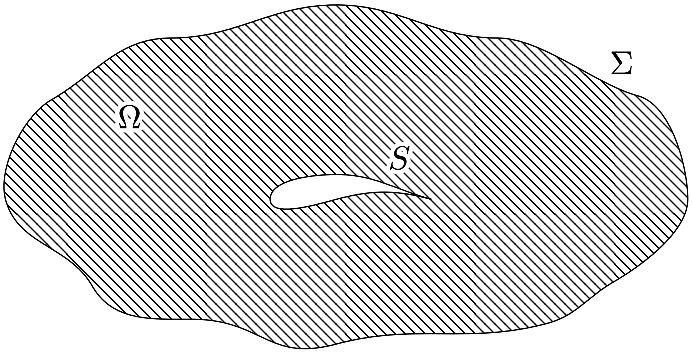


Figure 1: A cross section of Ω, illustrating the boundaries $S \cup \Sigma = \partial\Omega$.

The convergence of integrals of the form (2.8) only makes sense for an unbounded fluid domain should the differential form $\mathfrak{u}$ evanesce sufficiently fast. This is clear from the condition that $\mathfrak{u}$ shall have compact support. Such liberties are not ones we may always take, so we will make a concession in our definition of the manifold Ω. Suppose $\partial\Omega = S \cup \Sigma$, such that Σ encloses Ω, whilst S excises it, as is shown in figure 1. We may now extend Σ in such a way that the general properties of $\mathfrak{u}$ are conserved, now only requiring sufficiently rapid evanescence. In general, we shall not need such a strict definition of our manifold, as the rapid evanescence is sufficient for our purposes for Ω to be defined merely through the excision of the surface S from the ambient Euclidean space. An interesting inquiry to propose is of course how a punctured manifold behaves differently than one which is not. The answer is simple—problems will arise concerning the behavior of closed differential forms. We consider the following.

Lemma 2.10 (Poincaré). *Let $\mathfrak{a} \in \Lambda^k\mathbb{R}^n$ be an arbitrary closed form, meaning $\mathrm{d}\mathfrak{a} = 0$. There then exists a form $\mathfrak{b} \in \Lambda^{k-1}\mathbb{R}^n$ such that $\mathfrak{a} = \mathrm{d}\mathfrak{b}$.*

The terminology for a form being the exterior derivative of some other form is *exact*. The Poincaré lemma states that for the manifold of figure 1, we can always find some local representation such that any closed form is exact.

As was noted above, the Gauss–Ostrogradsky theorem of equation (2.1a) is a special case of equation (2.8). To show this, we recall the result of equation (2.7), and consider the following equality.

$$\int_\Omega \mathrm{d}(\boldsymbol{u} \lrcorner \mathrm{d}V) = \int_{\partial\Omega} \boldsymbol{u} \lrcorner \mathrm{d}V.$$

Indeed, $\boldsymbol{u} \cdot \hat{\boldsymbol{n}}\mathrm{d}S = u\,\mathrm{d}y \wedge \mathrm{d}z - v\,\mathrm{d}z \wedge \mathrm{d}x + w\,\mathrm{d}x \wedge \mathrm{d}y = \boldsymbol{u} \lrcorner \mathrm{d}V$. And with that, the major requisite of the boundary element method is proved—namely its progenitor. It will be shown that the boundary element method applies to solenoidal fluid flow of no vorticity.

Definition 2.11 (Hodge–Laplace operator). Let $\mathfrak{u} \in \Lambda^k\Omega$. We define the Hodge Laplacian to be

$$\Delta\mathfrak{u} = \text{đ}\mathrm{d}\mathfrak{u} + \mathrm{d}\text{đ}\mathfrak{u}. \tag{2.9}$$

Should $\Delta\mathfrak{u} = 0$, then $\mathfrak{u}$ is said to be harmonic.

Furthermore, we introduce a theorem of as great importance as the fundamental theorem of multivariate calculus for the study of fluid flow. It shall become evident that the properties of exactness and coëxactness of a differential form is invaluable information for the study of differential equations.

Theorem 2.12 (Hodge decomposition). *Any differential form $\mathfrak{u}$ may be decomposed as the sum $\mathrm{d}\mathfrak{a} + \text{đ}\mathfrak{b} + \mathfrak{c}$, where $\mathfrak{c}$ is harmonic.*

The Hodge–Laplace operator will be greatly studied in the forthcoming sections, as it is essential to the study of fluid flows. Also the study of harmonic functions, whose Laplacian is zero, is of significant importance—especially with regards to the study of functions of a complex variable, which we shall undertake promptly.

2.2 Complex analysis

An unbounded fluid, whose flow is invariant in a plane, will obviously be perfectly described by considering only the directions in which the fluid does vary. Examples of such flow are the cross section of an infinitely long cylinder, or any plane bisection a finite axisymmetric shape aligned with the flow. In any case, there are undoubtedly situations to which the application of solutions granted by consideration of two-dimensional problems to ones in apparently three dimensions is perfectly in order. It is such problems wherein the employment of complex analysis is of exceedingly great convenience. The formalism of the complex numbers may indeed be approached through the foundation laid in the previous section on exterior calculus.

> **Definition 2.13** (Almost complex manifold)**.** [1] Consider the real pseudo-RIEMANN manifold of even dimension Ω, and its tangent bundle $\mathrm{T}\,\Omega$. An almost complex structure is a skew-involute endomorphism J on the tangent space at any point in Ω, meaning $\mathrm{J}\circ\mathrm{J} = -\,\mathrm{id}$. A manifold equipped with such an almost complex structure is an almost complex manifold.

This definition is in many ways completely unhelpful to anyone not already familiar with the matter of complex numbers. First and foremost is the inclusion of demarkation that the manifold be *real* in some way. Indeed, Euclidean space upon which we base our notion of structure is complete, and indeed constructed by the real number line. One must surely agree when pressed that the field of real numbers is not algebraically closed, as the polynomial equation $x^2 + 1 = 0$, for instance, has no solution. Much like the rational numbers may be extended through irrational numbers, we extend the real numbers. Label then either one of the two solutions to this polynomial equation i, and consider the algebraic closure of the real numbers, the field of complex numbers $\mathbb{C}$. We shall in general denote an element therein $\mathfrak{z}$. Consider now $\mathbb{C}^n$, the complex space of n-tuples of complex numbers $\boldsymbol{\mathfrak{z}} = (\mathfrak{z}^1, \dots, \mathfrak{z}^n)$. It is clear that by writing that for any index k,

$$\mathfrak{z}^k = x^k + iz^k, \qquad x^k, z^k \in \mathbb{R},$$

there is a correspondence between the complex numbers and the plane. That is, there is an isomorphism $\mathbb{R}^2 \simeq \mathbb{C}$.

Second, is the elusiveness of the complex structure J. Extending the tangent bundle through complexification

The complex structure then takes the form

$$\mathrm{J}\partial_{x^k} = \partial_{z^k}, \qquad \mathrm{J}\partial_{z^k} = -\partial_{x^k} \tag{2.10a, b}$$

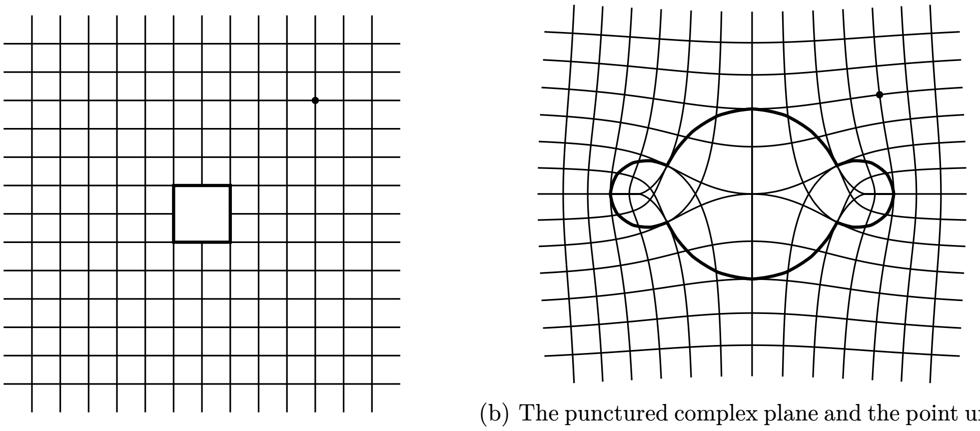
The complex conjugate is given by $\mathfrak{z}^* = x - iz$. $\mathrm{d}\mathfrak{z} = \mathrm{d}x + i\mathrm{d}z$.

$$\partial_{\mathfrak{z}} = {}^1\!/\!_2(\partial_x - i\partial_z), \qquad \partial_{\mathfrak{z}^*} = {}^1\!/\!_2(\partial_x + i\partial_z) \tag{2.11a, b}$$

2.3 Geometric fluid mechanics

In our effort to mathematically describe natural fluid flow, our work shall unsurprisingly be inspired by the principles of nature. The discussion that follows will elucidate to which extent our modeling actually reflects real fluids. Studying the physics of fluids is in large part shared by the consideration of the forces applied—dynamics—and the resulting motion—kinematics. It is the fluid kinematics we shall study first. Regarding the motion of the fluid, we should like to distinguish two frames of reference, namely the material and the spatial. The material frame of reference follows some specific fluid particle, whilst the spatial frame observes the fluid passing through some fixed point.[2] Consider some particle $\boldsymbol{x}_0 \in \Omega_0$, which is carried by some flow onto the point $\boldsymbol{x} \in \Omega$ at time t. Distinguishing Ω_0 is a formality—for our purposes it is the very same manifold as we otherwise consider, and the flow is an automorphism. This configuration describes nothing less than the dynamical system $\boldsymbol{x} = \boldsymbol{c}(\boldsymbol{x}_0, t)$, $\boldsymbol{x}_0 = \boldsymbol{c}(\boldsymbol{x}_0, 0)$, a solution of which is guaranteed locally by the PICARD–LINDELÖF theorem.[3] Clearly the velocity of the fluid at such a fixed point x must depend on the motion of the fluid particle. The nature of his dependency is however not immediately apparent. To this end, we consider the integral curves of the fluid velocity vector field $\boldsymbol{u}$ charted by $\boldsymbol{x}$. Such an integral curve we may in general label $\boldsymbol{c}(t)$, and are defined by its derivative coïnciding with the velocity field at each point. These are not necessarily constrained by an initial condition, but do correspond with the flow when they are. Although we shall not do so, it

[1] [50] KOBAYASHI & NOMIZU, p.121
[2] [77] TRUESDELL, §14, pp.29–35
[3] [3] ARNOL'D, §7.3, p.50



(a) The punctured complex plane and a point.

(b) The punctured complex plane and the point under a conformal map.

Figure 2: Both (a) and (b) show isolines of abscissae and ordinates of the complex plane punctured at the origin by a square. Under the conformal Žukovskij map of equation (3.1), the square becomes a quadrifoliate curve.

may be shown that flows actually constitute a diffeomorphism group $\mathscr{G}$, a group of differentiable flows whose inverses exist and are differentiable. Upon inspection, the group action reveals itself to be map composition. The flow from some initial time by some increment of time will obviously constitute the same chronology when flowing therefrom to some other increment of time, should the flow start from the very same commencing position and be carried to the sum of the two considered time increments. We conclude that tracing the same path of the flow backwards constitutes applying the inverse flow map, yielding the composition of the flow with the inverse flow, being the identity operator. We label the derivative of the flow with respect to time the particle velocity $\boldsymbol{u_c}$, representing the material frame of reference. Fixing some $\boldsymbol{x}$, it is now clear that the velocity in the spatial frame of reference is given by the expression $\boldsymbol{u} = \boldsymbol{u_c} \circ \boldsymbol{c}^{-1}(\boldsymbol{x})$. Should an explicit formula for this relationship be established, equations of motion may be derived from Newtonian principles and deliberation of dynamics, as we would possess the constitutive acceleration. We shall, however, explore another avenue through Hamiltonian mechanics. Considerable amount of such work can be done from very minimal physical considerations, as is shown in Vladimir Arnol'd' s works.[1,2] Although the results of these works are magnificent, the process by which the are derived are beyond the scope of this work, and indeed the writer at the present time. We shall therefore draw from other monographs as sources on connexions and symplectic forms to present these results. Thus we cannot motivate all steps in the derivations to come. The geometric formulation of hydrodynamics developed in the periphery of these named works of Arnol'd do indeed build upon the Hamiltonian formalism using the theory of groups mentioned above. They consider variations of fluid paths, and seek to minimize their effort by means of observing the extrema of the so-called *action*.

Another, perhaps more abstracted, albeit mathematically equivalent, perspective is considering a Lie–Poisson system, expounding upon the aforementioned diffeomorphism group rather directly. Should this group $\mathscr{G}$ consist of volumepreserving diffeomorphisms, it is denoted $\mathscr{G}_{\text{vol}}$. The kinetic energy is right invariant on the cotangent space $\mathrm{T}^*\mathscr{G}_{\text{vol}}$, yielding a Hamiltonian system on the Poisson manifold $\mathscr{X}^*_{\text{vol}}$, being the space of solenoidal vector fields tangent to $\partial\Omega$.[3] Such are the exact properties an ideal fluid flow ought to have, thus providing one with an *ad hoc* intuition on what the nature of ideal fluids are. We shall briefly define a connexion, followed by a theorem for illustrative purposes.

Definition 2.14 (Levi-Civita connexion)**.** Let $\boldsymbol{u} \in \mathrm{T}\,\Omega$ and $\mathfrak{a}$ be a section of some vector bundle. A connexion ∇ is a bilinear form. Should it preserve the metric in the sense that $\nabla\mathfrak{g} = 0$, and be torsion free, then it is a Levi-Civita connexion. These are determined by the Christoffel symbols

$$\nabla_i\partial_j = \Gamma^k_{ij}\partial_k, \qquad \Gamma^k_{ij} = {}^{\mathrm{g}^{k\ell}}\!/_2\big(\partial_j \mathrm{g}_{i\ell} + \partial_i \mathrm{g}_{j\ell} - \partial_\ell \mathrm{g}_{ij}\big). \tag{2.12}$$

Theorem 2.15 (Fundamental thoerem of pseudo-Riemannian geometry)**.** *Every Riemannian manifold has a unique Levi-Civita connexion.*

[1] [2] Arnol'd (1966)
[2] [4] Arnol'd & Hesin, ch.X
[3] [58] Marsden & Weinstein (1983)

A curve is called geodesic if $\nabla_{\dot{c}}\dot{c} = 0$. The insight of ARNOL'D leads to the satisfying conclusion that the EULER equation simply is a geodesic equation of the right-invariant L^2 metric on the diffeomorphism group, where the pressure is the constraining force orthogonal to the diffeomorphism group.[1,2] Upon adding a viscous diffusion term, we have the incompressible NAVIER–STOKES equation with no external forces,

$$\varrho\partial_t \mathfrak{u} + \varrho £_{\boldsymbol{u}}\mathfrak{u} = -\mathrm{d}p + \mu\Delta\mathfrak{u}, \qquad \text{đ}\mathfrak{u} = 0. \tag{2.13a, b}$$

The material derivative is here instead expressed as the pointwise derivative in time, in addition to the derivative with respect to the flow. We define it as follows.

Definition 2.16 (LIE derivative)**.** Let $\boldsymbol{u}$ be a vector field on the manifold Ω, and $\boldsymbol{c}$ the flow. For an arbitrary $\mathfrak{a} \in \Lambda^k\Omega$, the LIE derivative of $\mathfrak{a}$ by $\boldsymbol{u}$ is

$$£_{\boldsymbol{u}}\mathfrak{a} = \partial_t\big|_{t=0}\boldsymbol{c}^*\mathfrak{a} = \mathrm{d}(\boldsymbol{u} \lrcorner\, \mathfrak{a}) + \boldsymbol{u} \lrcorner\, \mathrm{d}\mathfrak{a}.$$

The vector field $\boldsymbol{u}$ is symplectic if $£_{\boldsymbol{u}}\mathfrak{a} = 0$ for a symplectic form $\mathfrak{a}$. For a vector field $\boldsymbol{v}$, the LIE derivative is defined to be

$$£_{\boldsymbol{u}}\boldsymbol{v} = [\boldsymbol{u}, \boldsymbol{v}].$$

We end the discussion on the nature of fluids as it pertains to this work with a rather illuminating theorem.

Theorem 2.17 (REYNOLDS transport)**.** *Let $\boldsymbol{u}$ be the vector field generated by the diffeömorphism $\boldsymbol{c}$ on the manifold $\Omega(t) = \boldsymbol{c}_t(\Omega_0)$. Let furthermore $\mathfrak{a}(t) \in \Lambda^k\Omega(t)$.*

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_\Omega \mathfrak{a} = \int_\Omega \partial_t\mathfrak{a} + £_{\boldsymbol{u}}\mathfrak{a}. \tag{2.14}$$

2.4 Potential fluid flow

The physical generation of lift is undoubtedly a viscous phenomenon, fully realized by the well known NAVIER–STOKES and continuity equations. As the measure of selfadhesion, viscosity naturally induces vortical motion in the fluid. Yet, a quasipotential formulation in which the effects of viscosity are realized, without the resulting motion being vortical, is what we shall utilize henceforth. Assume then, that the fluid we investigate is inviscid, and that the fluid velocity is small such that the MACH number is small. Upon assuming conservative body forces, the NAVIER–STOKES equation then reduces to the EULER and incompressibility equations

$$\varrho\mathrm{D}_t\boldsymbol{u} = -\nabla p, \qquad \nabla\cdot\boldsymbol{u} = 0. \tag{2.15a, b}$$

A vector field satisfying (2.15b) is called solenoidal. From the HODGE decomposition theorem, we may express the fluid velocity field as the sum $\boldsymbol{u}^\flat = \mathrm{d}\Phi + \text{đ}\mathfrak{v}$. It is apparent that the curl of the EULER equation (2.15a) is zero when the fluid flow is irrotational, such that the vector field describing the fluid flow is harmonic.

$$\Delta\Phi = 0, \qquad \boldsymbol{x} \in \Omega \tag{2.16}$$

This representation of the fluid is manageable and allows for powerful tools with which to investigate it, of which we shall promptly become aware. For we shall no longer consider the differential equation above, but instead an integral formulation more fitting for computeraided solving.

Consider two harmonic scalar functions ϕ and φ. Since all zero-forms are coclosed,[3] the HODGE–LAPLACE operator acting on zero-forms is equivalent to the familiar definition $\Delta\phi = \operatorname{div}\operatorname{grad}\phi$ of vector calculus. Consider furthermore the expressions $\text{đ}\phi\mathrm{d}\varphi$ and $\text{đ}\varphi\mathrm{d}\phi$, the codifferential working on the product. We take first the former, and expand the codifferential into its constituent stars and differential. From the definition of the HODGE star, we know it to be impotent acting on a scalar field. Thus,

$$\text{đ}\phi\mathrm{d}\varphi = \star\langle\mathrm{d}\phi, \mathrm{d}\varphi\rangle\,\mathrm{d}V + \phi\Delta\varphi = \star\langle\mathrm{d}\varphi, \mathrm{d}\phi\rangle\,\mathrm{d}V = \text{đ}\varphi\mathrm{d}\phi,$$

[1] [40] HESIN & MODIN (2025), Remark 2.3

[2] [45] IZOSIMOV & HESIN (2024)

[3] The proof is trivial. $\text{đ}\phi = \star\mathrm{d}\star\phi = \mathrm{d}\phi \wedge \mathrm{d}V$, since the volume form is closed. The grade of $\text{đ}\phi$ is then not congruent with the dimension of the manifold, and thus vanishes.

following from the symmetry of the inner product. Indeed, upon integrating their difference over the domain on which they are harmonic, we are free to exploit the fundamental theorem of multivariate calculus. Remarkably, the behavior of harmonic functions is entirely determined by their boundary data. This is in the special case of harmonic functions what is referred to as GREEN's second identity, which in the notation of vector calculus, can be denoted

$$\int_{\partial\Omega} (\phi\nabla\varphi - \varphi\nabla\phi) \cdot \hat{\boldsymbol{n}} \, \mathrm{d}S = 0, \tag{2.17}$$

Constructing now a GREEN function defined to be harmonic everywhere, with the exception at some point $\boldsymbol{ж} \in \mathcal{Q}$, we may recast equation (2.17) in terms of the CAUCHY principal value.[1] For the case of three dimensional flow, the GREEN function is given by

$$\mathrm{G}(\boldsymbol{x}; \boldsymbol{ж}) = \frac{1}{4\pi r}, \qquad r = |\boldsymbol{x} - \boldsymbol{ж}|. \tag{2.18a, b}$$

We confirm that $\Delta\,\mathrm{G}(\boldsymbol{x}; \boldsymbol{ж}) = \delta(\boldsymbol{x} - \boldsymbol{ж})$, where δ is the DIRAC delta function, meaning the GREEN function of equation (2.18a) is harmonic almost everywhere. We may insert this function for φ in equation (2.17), with the appropriate adjustments to the right-hand side. By integrating around the singularity at $\boldsymbol{ж}$, we find that

$$\mathrm{pv} \int_{\partial\Omega} (\phi\nabla_{\hat{\boldsymbol{n}}}\,\mathrm{G} - \mathrm{G}\,\nabla_{\hat{\boldsymbol{n}}}\phi) \, \mathrm{d}S = c\phi(\boldsymbol{ж}), \tag{2.19}$$

where c is some constant dependent on the location of $\boldsymbol{ж}$. For the directional derivative, we mean $\nabla_{\hat{\boldsymbol{n}}} = \hat{\boldsymbol{n}} \cdot \nabla$. This is the basis for the boundary element method, the numerical implementation of which this thesis seeks to employ.

2.5 Vortical fluid flow

Classical aerodynamics will study the effects of vortical motion in a fluid otherwise bereft thereof. A thorough exposition of that matter shall be handled in a later section, though the discussion of the influence vortices on fluids in general shall be accounted for promptly. We shall in this section examine how the BIOT–SAVART equation arises as a solution to the problem of a completely vortical fluid. For this purpose, we define the vorticity $\mathfrak{w} = \star\mathrm{d}\mathfrak{u}$, with the relation $\mathfrak{w}^\sharp = \boldsymbol{w}$. Should the fluid be otherwise nonvortical, we restrict the vorticity to the boundary. For discussions to be reëngaged at a later point in the work, we treat with some degree of rigor the matter of how such restrictions are mathematically founded.

We consider the vorticity equation, obtained by taking the curl of the NAVIER–STOKES equation (2.13a).

$$\varrho\partial_t \star\mathfrak{w} + \varrho\pounds_{\boldsymbol{u}} \star\mathfrak{w} = \mu\Delta\star\mathfrak{w}, \qquad \mathrm{đ}\mathfrak{w} = 0. \tag{2.20a, b}$$

Confer with BATCHELOR,[2] MARSDEN & WEINSTEIN,[3] or MODIN *et al.*[4] The question at hand is whether the vorticity is useful in the representation of the fluid flow we are interested in modeling. For any structure interacting with a fluid inducing a perturbed velocity, this velocity ought to not propagate indefinitely. In other words, it should be evanescent, a condition of regularity. Such vector fields may be described by decomposition into constituent HELMHOLTZ components, wherein

$$\boldsymbol{u} = \nabla\Phi + \nabla\times\boldsymbol{\Psi}, \qquad \boldsymbol{\Psi} = \boldsymbol{x}\chi. \tag{2.21a, b}$$

In terms of the vernacular of differential forms, the so-called scalar potential Φ is a zero-form, and the vector potential $\boldsymbol{\Psi}$ is a sharped one-form. Indeed, as the curl operator is simply $\star\mathrm{d}$, any choice of bivector $\mathfrak{v}$ such that $\star\mathfrak{v} = \Psi_i\,\mathrm{d}x^i$ will be approproiate, revealing the familiar HODGE decomposition of theorem (2.12). Such a choice of representation of the vector potential is called a gauge, and the gauge chosen in equation (2.21b) is that of GUMEROV & DURAISWAMI,[5] where we impose the condition that $\operatorname{div}\boldsymbol{\Psi} = 0$. In other words, the bivector $\mathfrak{v}$ is exact. It ought to be noted that the condition of evanescence is not one which applies to the HELMHOLTZ decomposition, but rather that of the convergence of integrals which will be discussed promptly.

Should the fluid happen to interact with a structure, it is only near the surface the effects of viscosity are apparent. A formulation, then, wherein the vorticity is nonzero only on the boundary of the object,

[1] explain
[2] [7] BATCHELOR, §5.2, pp.267–273
[3] [58] MARSDEN & WEINSTEIN (1983), eq.V
[4] [61] MODIN *et al.* (2010)
[5] [34] GUMEROV & DURAISWAMI (2006)

we may be able to proceed with a potential flow formulation in the rest of the fluid domain. Nevertheless, the laplacian term in the NAVIER–STOKES equation may be expanded in terms of equation (2.9) so as to reveal that

$$\mu\Delta\boldsymbol{u} = \mu\nabla(\nabla\cdot\boldsymbol{u}) - \mu\nabla\times\boldsymbol{w}. \tag{2.22}$$

This immediately yields the biharmonic equation for the potential function Φ when the vorticity is zero, which has been discussed by GUMEROV and DURAISWAMI in terms of implementation through fast multipole methods.[1] For we know the divergence of the velocity field to be zero, so that the Laplacian is equal to the curl of the vorticity. The connexion is immediately apparent between that of viscous effects encoded in the Laplacian term, and the vorticity of the fluid. Thus our motivation is plainly laid out before us—we wish to imitate the viscous effects that the fluid exhibits through a potential formulation. Employing the above HELMHOLTZ decomposition of equation (2.21), we find that by taking the laplacian thereof and substituting in the value of the vorticity of equation (2.22),

$$-\nabla\times\boldsymbol{w} = \nabla\times\Delta\boldsymbol{\Psi}.$$

We are free to equate the two arguments of the curl so that the vorticity equals the negative laplacian of the vector field $\boldsymbol{\Psi}$, or in terms of differential forms, $\Delta\star\mathfrak{v} = -\mathfrak{w}$. This is a POISSON equation, which is solved by the convolution $\star\mathfrak{v} = -\mathfrak{w} * \mathrm{G}$,[2] where G is the familiar GREEN function, expressed in local coördinates as

$$\Psi_i(\boldsymbol{x}) = -\int_{\Omega(\boldsymbol{\varkappa})} w_i\,\mathrm{G}\;\mathrm{d}V, \qquad \mathrm{G}(\boldsymbol{x},\boldsymbol{\varkappa}) = \frac{1}{4\pi|\boldsymbol{x}-\boldsymbol{\varkappa}|}.$$

Since it is the velocity $\mathfrak{u} = \operatorname{div}\mathfrak{v}$ we wish to find, we may simply calculate it from the above expression by taking the curl. Note that integration is done with respect to the variable $\boldsymbol{\varkappa}$, whilst the exterior derivative will be taken with respect to $\boldsymbol{x}$. To that end, we have $\partial_i\mathfrak{w}(\boldsymbol{\varkappa})\mathrm{d}x^i = 0$, so that by the graded product rule, we only differentiate the GREEN function. We then find the velocity vector field in terms of the vorticity, such that

$$\mathfrak{u}(\boldsymbol{x}) = -\int_\Omega \star\mathrm{d}(\mathfrak{w}\,\mathrm{G})\,\mathrm{d}V = -\int_\Omega \star(\mathfrak{w}\wedge\mathrm{d}\,\mathrm{G}\,)\,\mathrm{d}V. \tag{2.23}$$

The above expression seems to hold water,[3] as a similar result was obtained by HESIN.[4] The proof in the latter is somewhat different and more involved, as the vorticity is defined to be the two-form $\star\mathfrak{w}$, yielding a componentwise POISSON equation $\Delta u^i = \star(\mathrm{d}x^i\wedge\star\bar{\mathrm{d}}\star\mathfrak{w})$. This formulation results in the very same final equation, meaning the derivation, although novel to the present author, does not lend an unfamiliar approach to this common result. It does, however, lend credence to the result as valid in the formulation with which we have chosen to utilize. Barely predating that article is also a thorough investigaiton into vortex dynamics in fourdimensional space by SHASHIKANTH.[5] The reader already intimately familiar with the subject, will undoubtedly adduce the classic MARSDEN & WEINSTEIN article on incompressible fluids,[6] which does indeed mention—albeit not exactly analogous to the problem at hand—selfinducing velocity by a vortex filament in terms of the structures on the group of volumepreserving diffeömorphisms. Indeed, HESIN considers higherdimensional vortex filaments $C^{n-2}\subset\mathbb{R}^n$, referred to therein as membranes. We shall define strength of such a filament as follows.[7]

Definition 2.18 (Circulation)**.** Let C be a vortex filament, and D a transversal disk, intersecting C once. The strength of C is then given by

$$\Gamma = \int_D \iota_D^*\star\mathfrak{w} = \int_{\partial D}\iota_{\partial D}^*\mathfrak{u}.$$

We shall now explore the manner in which we may construct vorticity supported on such a vortex filament. We denote by ε the radius of the transversal disk D_ε enclosing the filament. Let the vortex filament be a line parametrized by the arclength s, such that the unit tangent vector is given by $\boldsymbol{t}(s) = \boldsymbol{x}'(s)$, an element of the tangent space on the filament. The following procedure is perhaps reminiscent of classical

[1] [34] GUMEROV & DURAISWAMI (2006)
[2] [76] TAYLOR, §3.4
[3] [78] TYSSVANG (2025)
[4] [39] HESIN (2012)
[5] [74] SHASHIKANTH (2012), §IV
[6] [58] MARSDEN & WEINSTEIN (1983)
[7] [74] SHASHIKANTH (2012), def.3.1

treatments of the matter of vortex tubes.[1] It is clear from definition 2.18 that the circulation is the very same as the flux of vorticity through the disk transversal to the filament. Upon the filament we construct an orthonormal coördinate system wherein we define a diffeömorphism from the parameter space and the the disk, onto the tube T_ε enclosing the filament, requiring the radius of the disk to be small. Thus, the tube on which the vorticity shall have support must have a cross-sectional area that is nonzero. We must have that at any point on the line the diffeömorphism maps identically in the parameter space onto $\boldsymbol{x}(s)$, and that the pushforward of the partial derivative ∂_s must be the tangent vector $\boldsymbol{t}(s)$. Thus, we define the local coördinate functions n^i on D_ε to be such that $\mathrm{d}n^i(\partial_{n^j}) = \delta^i_j$, their values being zero coïnciding with the filament. On the filament, the interior product of the tangent vector and the Euclidean volume form must then be the area form on the normal plane on the filament, $\boldsymbol{t} \lrcorner \mathrm{d}V = \mathrm{d}n^1 \wedge \mathrm{d}n^2$. This coördinate system is defined such that there is no intrinsic connexion between the vorticity and the radius of the disk. Clearly the above definition of the circulation makes no distinction of how great the readius of the disk shall be, and so the support of the vorticity shall need to be somehow dependent on the disk radius. For this, we introduce the mollifier φ, in the sense that $\varphi \in C^\infty_\mathrm{c}(\mathbb{R}^2)$, and $\varphi \geq 0$, with compact support on the unit disk. We regularize it such that

$$\int_{\mathbb{R}^2} \varphi(n^i)\,\mathrm{d}n^1 \wedge \mathrm{d}n^2 = 1. \tag{2.24}$$

The regularization of equation (2.24) admits $\varphi_\varepsilon = \varepsilon^{-2}\varphi({n^i}/{\varepsilon})$, now with support on D_ε.[2] We can now define locally the vorticity two-form and associated one-form, with compact support on the disk,

$$\star\mathfrak{w}_\varepsilon \equiv \Gamma\varphi_\varepsilon \boldsymbol{t} \lrcorner \mathrm{d}V, \qquad \mathfrak{w}_\varepsilon = \Gamma\varphi_\varepsilon \boldsymbol{t}^\flat, \tag{2.25}$$

when pulled back. As we noted earlier, at the centerline of tube coïnciding with the filament, the pushforward of the tangent partial derivative to the ambient space is the very tangent vector $\boldsymbol{t}$. Away from the center, however, this is not necessarily true, and the difference is $O(|n^i|) \leq O(\varepsilon)$. Let now s vary such that the disk $D_\varepsilon(s)$ is the corresponding transversal disk thereat, as in definition 2.18. We then have

$$\int_{D_\varepsilon(s)} \iota^*_{D_\varepsilon(s)} \star\mathfrak{w}_\varepsilon = \Gamma \int_{D_\varepsilon} \varphi_\varepsilon \,\mathrm{d}n^1 \wedge \mathrm{d}n^2 = \Gamma, \tag{2.26}$$

having pulled back the vorticity two-form using the diffeömorphism. Evidently, the vorticity flux along the length of the filament is constant, equal to Γ. We shall now see that the supported vorticity $\star\mathfrak{w}_\varepsilon$ converges in a weak sense to a current supported on the filament, which is sufficient for our purposes. To that end, we let $\mathfrak{a}$ be a one-form to test against, and consider the L^2 inner product $(\mathfrak{a}, \mathfrak{w}_\varepsilon)_\Omega$. We have that for small ε, the support of $\star\mathfrak{w}_\varepsilon$ is contained in T_ε, so that the domain of integration is restricted to the tubular neighborhood around the filament. The integrand is then the pullback of the expression $\mathfrak{a} \wedge \star\mathfrak{w}_\varepsilon$. That is,

$$(\mathfrak{a}, \mathfrak{w}_\varepsilon)_\Omega = \int_\Omega \mathfrak{a} \wedge \star\mathfrak{w}_\varepsilon = \Gamma \int_C \boldsymbol{t} \lrcorner \mathfrak{a}_{\boldsymbol{x}(s)} \,\mathrm{d}s + O(\varepsilon). \tag{2.27}$$

This is essentially identical to the reuslt presented by Hesin,[3] stating that vorticity two-form is a δ-function with support on the filament. The procedure is not as crude as simply inserting the vorticity one-form in the above Biot–Savart equation, for it converges weakly. We must now consider $\mathfrak{u}_\varepsilon$, and the limit as the radius evanesces.

$$\mathfrak{u}_\varepsilon(\boldsymbol{\varkappa}) = -\int_\Omega \star(\mathfrak{w}_\varepsilon \wedge \mathrm{d}\,\mathrm{G}\,)\,\mathrm{d}V = -\Gamma \int_C \int_{D_\varepsilon} \star(\boldsymbol{t}^\flat \wedge \mathrm{d}\,\mathrm{G}\,)\varphi_\varepsilon \,\mathrm{d}n^1 \mathrm{d}n^2 \,\mathrm{d}s. \tag{2.28}$$

Fixing a point $\boldsymbol{\varkappa} \notin C$, it is clear that the kernel is smooth, and that we may perform the limit process. Finally, we have the Biot–Savart formulation of the induced velocity field[4,5]

$$\boldsymbol{u}(\boldsymbol{\varkappa}) = \frac{\Gamma}{4\pi} \int_C \frac{\boldsymbol{x} - \boldsymbol{\varkappa}}{r^3} \times \mathrm{d}\boldsymbol{s}. \tag{2.29}$$

The great ambition of this subsection was to derive the vortex filament formulation of the Biot–Savart equation in a novel manner, employing the tools of differential geometry. This was unfortunately thwarted

[1] [77] Truesdell, §43, pp.82–84
[2] [69] de Rham, §15, pp.61–70
[3] [39] Hesin (2012), §§2.2, 6.1
[4] [21] Deng *et al.* (2020), eq.6
[5] [72] Schlichting & Truckenbrodt, p.98, eq.2.161

by greater minds at times greatly preceeding it, and as such it only serves as a point of comparison of the mastery of these tools. Furthermore of interest is the apparent ease with which the argumentation above is guided through the language of currents. Indeed, a filament is nothing more than a circulation weighted current acting on test one-forms.[1] We close this section with a rather interesting interpretation of the theory of currents, stating that no vortex filament can terminate independently of boundary data or other filaments, forming a vortex ring, essentially reframing the theorems of HELMHOLTZ. A filament current with no initial endpoint contribution, can have no such contribution at a later time. By the theory already established, we may state this succinctly in the following manner.

> **Theorem 2.19** (The circulation theorem)**.** *Consider an inviscid fluid, such that the righthand side of (2.20a) is zero. By theorem 2.17, there can be no change in circulation.*

3 Aerodynamics

3.1 Boundary layers

It has been shown that for at least moderate REYNOLDS numbers, the fluid flow over an airfoil in two dimensions only depends on the curvature as a first order correction.[2]

3.2 Airfoil theory

Any blade of a wind turbine is composed from root to tip of a smooth distribution of cross sections, much like the one illustrated in figure 4a. Referred to as airfoils, these cross sections are optimized for the generation of lift. As was alluded to in the section on complex analysis, these are wholly two dimensional objects, whose ability to generate lift can be quantified by the means of this theory. Their mathematical representation warrants discussion, as it lends itself to the impending numerical analysis by way of comparison to theoretically derived values. Consider then the complex variable $\mathfrak{z}' = x' + iz'$ whose values trace the circle centered at the origin of radius c. The ŽUKVOSKIJ transform takes a complex value and maps it to the point[3]

$$\mathfrak{z} = f(\mathfrak{z}') = \mathfrak{z}' + \frac{c^2}{\mathfrak{z}'}. \tag{3.1}$$

Consider concentric circles about the origin, whose abscissae and ordinals are for any radial value are given respectively by $x' = r\cos\theta$ and $z' = r\sin\theta$. Under the image of the ŽUKOVSKIJ map, these circles are of the form

$$x = \left(r + \frac{c^2}{r}\right)\cos\theta, \qquad z = \left(r - \frac{c^2}{r}\right)\sin\theta. \tag{3.2a, b}$$

We shall show that these constitute confocal ellipses in the image domain. Fixing the radius, and squaring each of the equations of (3.2), yields by the Pythagorean identity $\sin^2\theta + \cos^2\theta = 1$ the relations

$$\frac{x^2}{a^2} + \frac{z^2}{b^2} = 1, \qquad a = r + \frac{c^2}{r}, \qquad b = \left| r - \frac{c^2}{r} \right|. \tag{3.3a, b, c}$$

Equation (3.3a) does indeed constitute the equation of an ellipse with foci $\pm 2c$, wherein a is the major semiaxis, and b the minor semiaxis. Fixing instead the angular variable in equations (3.2) corresponds to rays emanating from the origin. Much the same procedure yields an expression in terms of the difference of squares in

$$\frac{x^2}{a'^2} - \frac{y^2}{b'^2} = 1, \qquad a' = 2c\cos\theta, \qquad b' = 2c\sin\theta \tag{3.4a, b, c}$$

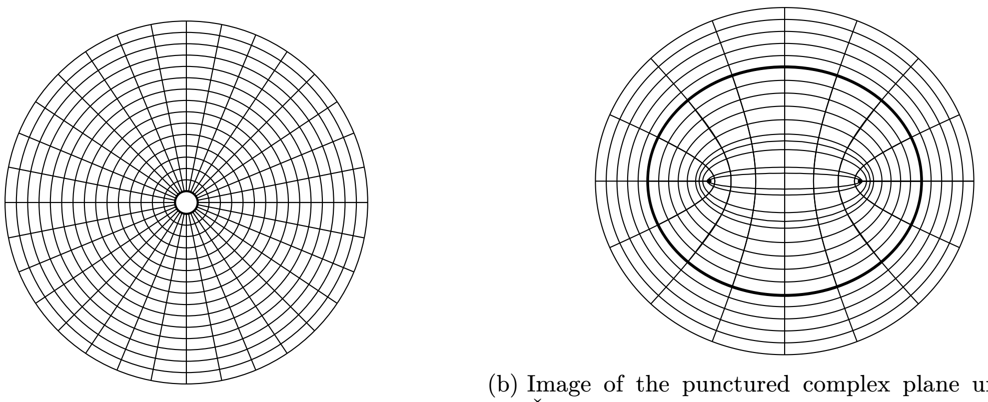
Equation (3.4) is the equation for the hyperbola with semiaxes a' and b', confocal to the above ellipses. Such mapping of concentric circles to confocal ellipses is illustrated explicitly in figure 3. A more visually pleasant, and perhaps more enlightening presentation of the ŽUKOVSKIJ transform is found in terms of the complex variable defined by the relation

$$\mathfrak{z}' = ce^{\text{з}}, \qquad \text{з} = \text{ж} + i\text{ч}. \tag{3.5a, b}$$

[1][69] DE RHAM

[2][52] LEDERHILGER (2024)

[3][72] SCHLICHTING & TRUCKENBRODT, p.404, eq.6.37



(a) Polar representation of the punctured complex plane.

(b) Image of the punctured complex plane under the Žukovskij transform.

Figure 3: Concentric circles are mapped to confocal ellipses. Rays are mapped to hyperbolae.

By simple calculation through the rules of arithmetic of the natural logarithm, the real and imaginary parts of this new complex variable are found to respectively be $\text{ж} = \ln {}^{r}/_{c}$ and $\text{ч} = \theta$. Thus we find that the Žukovskij transform of equation (3.1) may be equivalently cast as the map

$$\mathfrak{z} = 2c\cosh \text{з}, \qquad x = 2c\cosh \text{ж} \cos \text{ч}, \qquad z = 2c \sinh \text{ж} \sin \text{ч}, \tag{3.6a, b, c}$$

though this will not be of much use for the purposes of our analysis except perhaps enlightening the behavior of the map. A more thorough treatment on the matter of elliptic coördinates is given in Kennard's monograph on irrotational fluids.[1] Nevertheless, the connexion between the transform at hand and the blades of a wind turbine is not these confocal ellipses from concentric circles of the origin. For shifting the center of a circle everso slightly away from the origin reveals the very airfoil of figure 4a. Consider a circle of radius a with center at the imaginary number s, whose perimeter intersects both the real numbers $\pm c$. Consider then the complex numbers $\mathfrak{z} \pm c$ originating from $\pm c$, such that they both intersect the circle. The angle at the point of incidence with the circle of the resulting triangle we name ϑ, it being the difference in argument of the two complex numbers $\mathfrak{z} \pm c$. Transforming each of these complex numbers using (3.1), we find that

$$\frac{\mathfrak{z} - 2c}{\mathfrak{z} + 2c} = \frac{(\mathfrak{z}' - c)^2}{(\mathfrak{z}' + c)^2}, \tag{3.7}$$

and thus $\vartheta' = 2\vartheta$, where ϑ' is the difference in argument of the transformed values. For any point of intersection, ϑ will remain constant,[2] so that the circle is mapped to a circular arc between the focal points $\pm 2c$ under the Žukovskij transform. Let the angle from either points $\pm c$ to the center s be named β, whose value is evidently given by $\beta = {}^{\pi}/_{2} - \vartheta$. By the above double angle relationship of the interior angle to its image under the Žukovskij transform, the angle between the tangent of the resulting circular arc and the chord is 2β. Denote the radius of the circle, whose segment constitutes the chord of the circular arc, A, and let the camber of the circular arc—the measure of its deviation from the chord—be denoted C. We then have the following.[3]

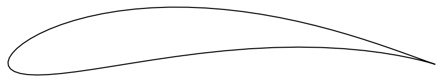
$$A = \frac{a^2}{\sqrt{a^2 - c^2}}, \qquad C = \frac{\tan\beta}{2} \tag{3.8}$$

Adding a real part to the center s will yield the airfoil shape we are after. Construct a circle whose perimeter coïncides with c and encompasses $-c$. Such airfoils have a cusp at the trailing edge, as is shown in figure ??. The airfoils of von Kármán & Trefftz are generalizations of those of Žukovskij, and modify the cusps to instead have a nonzero opening angle.[4] It can be shown that for $k = 2$, this is equivalent to the expression[5]

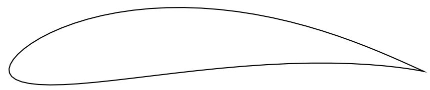
$$\frac{\mathfrak{z}}{kc} = \frac{(\mathfrak{z}' + c)^k + (\mathfrak{z}' - c)^k}{(\mathfrak{z}' + c)^k - (\mathfrak{z}' - c)^k}. \tag{3.9}$$

From this analytical expression, we may find the skeleton of the airfoil

[1] [49] Kennard, §§61,81–82
[2] [24] Euclid, prop.XXI, lib.iii, p.71
[3] [49] Kennard, p.178
[4] [68] Prandtl & Tietjens, §106, pp.182–185
[5] [60] Milne-Thomson, pp.125–128



(a) Žukovskij airfoil



(b) von Kármán–Trefftz airfoil

Figure 4: Two airfoils described respectively be equations (3.2) and (3.9), whose parameters are $c = 1.4$ and $s = {}^1/_4(i - {}^3/_5)$. The trailing edge parameter is $k = 1.94$.

3.3 Thin wing approximation

Until the conclusion of this work, we concern ourselves with the airfoil skeleton theory of Birnbaum, wherein the aerodynamical calculations are performed on the mean camber line only. For it shall be shown that the matter of flow due to thickness and inclination and camber are fundamentally different, and indeed separable. Consider an airfoil given as a closed curve in the complex plane, whose chord coïncides with the abscissal coördinate. The mean camber line is the curve that for any abscissal value lies exactly midway between the two ordinal values of the airfoil. It is evident that this curve describing the airfoil can then be depicted entirely by the sum of the mean camber line and the thickness function.

In most practical scenarios, one would not be given a Žukovskij transform with which to analytically determine the mean camber line. One would instead be given an airfoil in terms of discrete points along its contour. The crucial observation to be made is that these points serve as vertices for a polygon inscribed in the airfoil. If one were given a distribution of vertices such that the number along the suction side and pressure side were equal, one could consider the midpoint along the diagonal of corresponding vertices, numbered from leading edge to trailing edge, or *vice versa*, as proposed by Hess.[1] Connecting each such midpoint from leading edge to trailing edge would yield an acceptable approximation to the mean camber line. Should the number of provided vertices be different on the suction and pressure sides, however, the procedure through which to obtain the mean camber line is not as evident. We shall determine an approximation to the mean camber line through studying the medial axes of polygons. Further, we conjecture these to be the optimal approximations, for the error of the medial axis to the mean camber line cannot be larger than the error of the inscribed polygon relative to the true curve of the airfoil. The medial axis was conceived of by Blum in a selcouth passage of writing on biological and visual shape description.[2] Its definition is as follows.[3]

Definition 3.1 (Medial axis)**.** Given a simple polygon $\Omega^\dagger$, its medial axis is the set of points $\{\boldsymbol{x}_i\}$ internal to it, such that there are at least two points on its boundary that are equdistant from and are closest to $\{\boldsymbol{x}_i\}$.

The peculiar choice of notation of the polygon stems from the interior of the wing, whose cross section is the airfoil at hand, being the complement of the manifold inhabiting the fluid of interest. Forcing a simple polygon, we employ the Jordan curve theorem,[4] yielding a consistent notion of interior. We can then implement an algorithm to determine the Voronoj diagram, whose set of edges is a superset of the medial axis,[5] simply terminating branches thereof incident with reflex vertices.

Conjecture 3.2. *For an airfoil sampled at a finite number of points, the medial axis of the polygon thereof is the best approximation of its mean camber line.*

4 Spherical functions

It is perhaps self-evident that standard procedure for dealing with the calculations involving the Green function consists of expressing it in terms of spherical harmonics. The proverbial radius is in the very expression, after all. It is nevertheless the case that it shall be crucial in the development of the fast multipole method. We shall not digress completely into the group theory of the projective spaces of hyperspherical functions, but we shall lay some foundation to motivate our representation of functions in terms of spherical harmonics. To somewhat motivate our motivation to this end, we consider the points $\boldsymbol{x}$ and $\boldsymbol{\varkappa}$ on the d-dimensional manifold Ω, whose angle between them, of the triangle in the plane described by them and the origin, is denoted α. For although these points lie in d-dimensional space, we know them to form a unique plane, within which the angle α may be defined. The points themselves may

[1] [42] Hess (1974), see fig.7a
[2] [9] Blum (1967)
[3] [54] Li (1982),
[4] [29] Goluzin, pp.8&36
[5] [54] Li (1982), Corollary 3.

be given in terms of spherical coördinates, so that the distance between the points $\boldsymbol{x}$ and $\boldsymbol{\varkappa}$, may be written in terms of the law of cosines,[1]

$$r = \sqrt{|\boldsymbol{x}|^2 - 2|\boldsymbol{x}||\boldsymbol{\varkappa}|\cos\alpha + |\boldsymbol{\varkappa}|^2}. \tag{4.1}$$

Now the Green function is written

$$\mathrm{G}(\boldsymbol{x};\boldsymbol{\varkappa}) = \frac{c|\boldsymbol{\varkappa}|^{2-d}}{\left(1+\varepsilon^2-2\varepsilon\cos\alpha\right)^{\nu}}, \qquad \varepsilon = \frac{|\boldsymbol{x}|}{|\boldsymbol{\varkappa}|}, \qquad c = \frac{\Gamma(d/2+1)}{2\nu d\pi^{d/2}}. \tag{4.2a, b, c}$$

Here we have defined a new parameter $\nu = d/2 - 1$ for the sake of consistency with forthcoming notation. Concerning the constant term c in equation (4.2c), the reader is referred to a text on differential equations in higher dimensions—it encodes the volume of the unit ball in d dimensions, and reduces to the familiar reciprocal of 4π for the three-dimensional case. $\Gamma(\mathfrak{z})$ is the gamma function. Through a binomial series expansion, a power series of equation (4.2a) can be found, the coëfficients of which some authors then define a set of functions to ascribe.[2]

$$\mathrm{G}(\boldsymbol{x};\boldsymbol{\varkappa}) = c|\boldsymbol{\varkappa}|^{2-d}\sum_{k=0}^{\infty}\varepsilon^k\,\mathrm{C}_k^{\nu}(\cos\alpha), \quad \varepsilon < 1, \tag{4.3a, b}$$

Indeed, in terms of generating functions assuming the form f_k, so that[3]

$$g(R) = \sum_{k=0}^{\infty} a_k f_k(x)\varepsilon^k, \qquad R = \sqrt{1-2x\varepsilon+\varepsilon^2}, \tag{4.4a, b}$$

the so-called Gegenbauer ultraspherical functions generate functions of the form $g(R) = R^{-2\alpha}$.

4.1 Ultraspherical functions

We shall digress but for a brief moment to introduce functions upon spheres. A more thorough discussion of the topic immediately at hand is of course to be found in any monograph on the topic. The purpose of this digression is to clearly illustrate for what reasons and by which means we expand the Green function into representations in terms of spherical functions. Let ν be a positive semiïnteger, meaning that $2\nu \in \mathbb{N}^+$, and let n be a positive integer. The Gegenbauer ultraspherical polynomials $\mathrm{C}_n^{\nu}(x)$ then constitute a complete orthonormal basis for the L^2-space on the unit interval, with respect to the measure $\left(1-x^2\right)^{\nu-1/2}\,\mathrm{d}x$, indexed in n.[4] The significance of this measure is not immediately apparent. Alternatively to the above definition as generating functions, we have the definition through the hypergeometric function,[5]

$$\mathrm{C}_n^{\nu}(x) = \frac{\Gamma(2\nu+n)}{n!\Gamma(2\nu)}\,\mathrm{F}\left(2\nu+n,\ -n;\ \nu+1/2;\ 1/2 - x/2\right), \tag{4.5}$$

The hypergeometric function $\mathrm{F}(a,b;c;\mathfrak{z})$ is a special function, whose full treatment is far beyond the scope of this work. It warrants mentioning that it may either be defined in terms of a power series,[6] or more relevant for our purposes, as the unique real analytic of the hypergeometric equation

$$\mathfrak{z}(1-\mathfrak{z})\frac{\mathrm{d}^2 f}{\mathrm{d}\mathfrak{z}^2} + (c-(a+b+1)\mathfrak{z})\frac{\mathrm{d}f}{\mathrm{d}\mathfrak{z}} - abf = 0, \qquad |\mathfrak{z}| < 1. \tag{4.6a, b}$$

This equation will reappear in an impending subsection, as a direct consequence of the theory for which we seek a foundation. The ultraspherical functions are related to the associated Legendre spherical function through the following expression.[7]

$$\mathrm{C}_n^{\nu}(x) = \frac{\Gamma(\nu+1/2)\Gamma(2\nu+n)}{n!\Gamma(2\nu)}\left(\frac{x^2-1}{4}\right)^{\frac{1-2\nu}{4}}\mathrm{P}^{1/2-\nu}_{n+\nu-1/2}(x). \tag{4.7}$$

Through algebraic manipulations, the definition of the associated Legendre functions are found to be[8]

[1] [18] Collins & Osler (2011)
[2] [6] Avery, eq.3.5, p.26
[3] [1] Hochstrasser, §22.9, p.783
[4] [75] Takeuchi, p.222
[5] [75] Takeuchi, p.194
[6] [1] Oberhettinger, eq.15.1.1, p.556
[7] [1] Hochstrasser, eq.22.5.60, p.780
[8] [1] Stegun, eq.8.1.2, p.332

$$\mathrm{P}_n^m(\mathfrak{z}) = \frac{1}{\Gamma(1-m)}\left(\frac{\mathfrak{z}+1}{\mathfrak{z}-1}\right)^{m/2} \mathrm{F}\left(-n,\, n+1;\, 1-m;\, {}^1\!/\!_2 - {}^{\mathfrak{z}}\!/\!_2\right) \tag{4.8}$$

For $\nu = 1/2$, we have that $m = 0$, where $\mathrm{P}_k^0 = \mathrm{P}_k$ is the k^{th} LEGENDRE polynomial.[1]

4.2 Spherical coördinates

For a moment we consider the Cartesian coördinates x^i in $\mathbb{E}^{d+1}$, due to the ease of calculation and later discussion on hyperspherical operators. The parametrization of hyperspherical coördinates takes the implicit form[2]

$$x^j = r\cos\alpha^j \Pi_{j-1}, \qquad \Pi_n = \prod_{k=1}^{n} \sin\alpha^k; \tag{4.9a, b}$$

$$x^1 = r\cos\alpha^1, \qquad x^d = r\sin\alpha^0 \Pi_{d-1}, \qquad x^{d+1} = r\cos\alpha^0\Pi_{d-1}. \tag{4.10a, b, c}$$

where $j \in \{2, \dots, d-1\}$ for $d \geq 3$. Nevermind the indexing, as we shall reïndex at a later point. From the twodimensional construction of polar coördinates in α^0, upon defining $\Pi_0 = 1$, we may iteratively construct the coördinates of the hypersphere. These polar coördinates relate the abscissa and ordinate of a point in the plane to the angle α^0 between them, free to range from zero to 2π. Raising this point into threedimensional space, an azimuthal angle α^1 is necessarily introduced. It is evident that the position of the point is completely described when this azimuthal angle varies from zero to π. We thus have in threedimensional space $\boldsymbol{x} = (r\cos\alpha^1, r\sin\alpha^0\sin\alpha^1, r\cos\alpha^0\sin\alpha^1)$. Drawing forth ever more azimuthal angles in ever higher dimensions, the parametrization above becomes evident.

For such hyperspherical coördinates, we wish to derive the HODGE–LAPLACE operator for the analysis of harmonic functions. This we do by pulling back the Cartesian metric tensor in Euclidean space $\mathfrak{g}'$, effectively performing a coördinate transformation. Let the parametrization (4.9), (4.10) be given by ϕ, such that $\mathrm{g}_{ij} = (\phi^*\mathrm{g}')_{ij}$. Note that $x^k = r\upsilon^k$, where υ^k is the corresponding angular part of the parametrization. We know the Cartesian metric to be the identity matrix, so the spherical representation of the metric tensor $\mathfrak{g}$ can be simplified as follows.

$$\mathrm{g}_{ij} = \mathrm{g}'_{k\ell}\frac{\partial x^k}{\partial\alpha^i}\frac{\partial x^\ell}{\partial\alpha^j} = \frac{\partial x^k}{\partial\alpha^i}\frac{\partial x_k}{\partial\alpha^j} = r^2\frac{\partial\upsilon^k}{\partial\alpha^i}\frac{\partial\upsilon_k}{\partial\alpha^j} \tag{4.11}$$

The indexing is arbitrary, as we have not accounted for the entries concerning r and α^0 of equation (4.10), though it is costumary to place r at the first and α^0 at the last indices. To calculate the expression $\partial_i\upsilon^k$, we shall need to find the derivative $\partial_i\Pi_{k-1}$, of which we may identify two possibilities. Consider for the moment $\partial_i\Pi_j$ such that $j < 1$. It is evident that the expression does not depend on α^i, and that the result is zero. Conversely, for $j \geq i$, we differentiate the i^{th} term in the product (4.9b), amounting to multiplying by the cotangent. That is, $\partial_i\Pi_j = \cot\alpha^i\Pi_j$. Thus we have that $\partial_i\upsilon^k = \cos\alpha^k\partial_i\Pi_{k-1} - \sin\alpha^k\Pi_{k-1}\delta_i^k$, where $k \in \{1, \dots, d-1\}$, can be written

$$\partial_i\upsilon^k = \begin{cases} 0, & k < i \\ -\Pi_i, & k = i \\ \upsilon^k\cot\alpha^i, & k > i \end{cases}. \tag{4.12}$$

It is clear that we need not consider terms $k < j$ in the calculation of the metric tensor of equation (4.11). Using, then, the derivatives of (4.12), we find the offdiagonal terms to be

$$\mathrm{g}_{ij} = -r^2\upsilon^j\cot\alpha^i\Pi_j + r^2\cot\alpha^i\cot\alpha^j\sum_{k=j+1}^{d+1}\left(\upsilon^k\right)^2, \qquad i \neq j. \tag{4.13}$$

The rightmost sum in equation (4.13) we may evaluate now evaluate using the the following two facts. First, the EULER identity states that $\sin^2\alpha^\ell + \cos^2\alpha^\ell = 1$. Recall that according to equation (4.10b, c), $\upsilon^d = \sin\alpha^0\Pi_{d-1}$ and $\upsilon^{d+1} = \cos\alpha^0\Pi_{d-1}$. Second, we know $\Pi_{k-1} = \csc\alpha^k\Pi_k$. Thus,

$$\sum_{k=j+1}^{d+1}\left(\upsilon^k\right)^2 = \left(\Pi_{d-1}\right)^2 + \sum_{k=j+1}^{d-1}(1-\sin^2\alpha^k)\left(\Pi_{k-1}\right)^2 = \left(\Pi_{d-1}\right)^2 + \sum_{k=j+1}^{d-1}\left(\Pi_{k-1}\right)^2 - \left(\Pi_k\right)^2.$$

[1] [1] STEGUN, pp.332–353

[2] [10] BLUMENSON (1960)

The nature of the sum is clearly *telescoping*, subsequent terms in the series canceling preceding terms.[1] In fact, we the only term remaining is ${\Pi_j}^2$. The leftmost term in equation (4.13) is the case where $j = k$. This we also rewrite, using (4.9a), so that the offdiagonal entries of the metric tensor are

$$\mathrm{g}_{ij} = -r^2 \cot\alpha^j \cot\alpha^i {(\Pi_j)}^2 + r^2 \cot\alpha^i \cot\alpha^j {(\Pi_j)}^2 = 0, \qquad i \neq j.$$

We find that the first and last columns and rows of the matrix to also be zero, when accounting for r and α^0—only diagonal entries exist for the metric tensor. The interior diagonal entries are

$$\mathrm{g}_{ii} = r^2(1+\cot^2\alpha^i){(\Pi_i)}^2 = r^2 \csc^2\alpha^i {(\Pi_i)}^2 = r^2 \prod_{k=1}^{i-1} \sin^2\alpha^k. \tag{4.14}$$

As for the first and the last entries of the matrix, the radial term is easily remedied. Clearly $\partial_r x^k = \upsilon^k$, so that $\mathrm{g}_{rr} = \delta_{k\ell}\upsilon^k\upsilon^\ell = 1$. For nonzero values of k, the derivatives $\partial_0\upsilon^k$ are zero. We then have

$$\mathrm{g}_{00} = r^2\partial_0\upsilon^k\partial_0\upsilon^k = r^2{(\upsilon^d)}^2 + r^2{(\upsilon^{d+1})}^2 = r^2\prod_{k=1}^{d-1}\sin^2\alpha^k. \tag{4.15}$$

Assembling the full metric tensor, we have in matrix form

$$\mathfrak{g} = \begin{pmatrix} 1 & & & & \\ & \ddots & & & \\ & & r^2 \prod_{k=1}^{i-2} \sin^2\alpha^k & & \\ & & & \ddots & \\ & & & & r^2\prod_{k=1}^{d-1}\sin^2\alpha^k \end{pmatrix}, \qquad \sqrt{\det(\mathfrak{g})} = r^d \prod_{i=1}^{d-1} \sin^{d-i}\alpha^i. \tag{4.16a, b}$$

Note the recursive nature of the angular entries in the metric tensor (4.16a). For upon denoting by $\mathbb{S}^d \subset \mathbb{E}^{d+1}$ the angular subspace, it is clear that $\mathfrak{g}_{\mathbb{S}^d} = \mathrm{d}\alpha^1 \otimes \mathrm{d}\alpha^1 + \sin^2\alpha^1 \mathfrak{g}_{\mathbb{S}^{d-1}}$, with $\mathfrak{g}_{\mathbb{S}^1} = \mathrm{d}\alpha^0 \otimes \mathrm{d}\alpha^0$. In general, the submatrices along the diagonal follow the pattern

$$\mathfrak{g}_{\mathbb{S}^{d-j+1}} = \mathrm{d}\alpha^j \otimes \mathrm{d}\alpha^j + \sin^2\alpha^j \mathfrak{g}_{\mathbb{S}^{d-j}}, \qquad \sqrt{\det(\mathfrak{g}_{\mathbb{S}^{d-j+1}})} = \sin^{d-j}\alpha^j \sqrt{\det(\mathfrak{g}_{\mathbb{S}^{d-j}})}. \tag{4.17a, b}$$

We shall see that this is quite useful in the understanding of harmonic functions in hyperspherical coördinates.

4.3 Spherical harmonics

With the metric tensor at hand, we may express the Hodge–Laplace operator with respect to hyperspherical coördinates. For zero-forms, we know the operator to be $\eth\mathrm{d}f = -\star\mathrm{d}\star\mathrm{d}f$. We shall denote for brevity $\mathrm{d}V = \sqrt{\det(\mathfrak{g})}\mathfrak{a}$, such that $\star\mathrm{d}f = \sqrt{\det(\mathfrak{g})}\mathrm{g}^{ij}\partial_i f \partial_j \lrcorner\, \mathfrak{a}$. Clearly $\mathrm{d}(\partial_j \lrcorner\, \mathfrak{a}) = 0$, meaning

$$\mathrm{d}\star\mathrm{d}f = \mathrm{d}\Big(\sqrt{\det(\mathfrak{g})}\mathrm{g}^{ij}\partial_i f\Big) \wedge \partial_j \lrcorner\, \mathfrak{a} = \partial_k\Big(\sqrt{\det(\mathfrak{g})}\mathrm{g}^{ij}\partial_i f\Big)\mathrm{d}x^k \wedge \partial_j \lrcorner\, \mathfrak{a}.$$

We know $\mathrm{d}x^k \wedge \partial_j \lrcorner\, \mathfrak{a} = \delta^k_j\mathfrak{a}$, removing all terms of index $k \neq j$. Applying the Hodge star operator once more to the expression above, we get the following expression for the Hodge–Laplace operator in local coördinates.[2]

$$\Delta f = \frac{1}{\sqrt{\det(\mathfrak{g})}}\frac{\partial}{\partial x^j}\left(\sqrt{\det(\mathfrak{g})}\mathrm{g}^{ij}\frac{\partial f}{\partial x^i}\right) \tag{4.18}$$

It is known that the Hodge–Laplace operator on $\mathbb{E}^{d+1}$ may be decomposed so as to separate the dependence of the radial distance from the origin, and the placement of the point of evaluation on the unit

[1] [56] Lindstrøm, p.141

[2] [66] Nishikawa, p.116

hypersphere $\mathbb{S}^d$.[1] This is a digestible result once one considers the projection of a point from the origin, through the unit hypersphere, unto the designated radius. This is obviously something we can show using equation (4.16), but shall not do. Thus we have the following representation the HODGE–LAPLACE operator.

$$\Delta_{\mathbb{E}^{d+1}} f = r^{-d}\partial_r(r^d\partial_r f) + r^{-2}\Delta_{\mathbb{S}^d} f(\mathbb{S}(r)); \tag{4.19}$$

$$\Delta_{\mathbb{S}^j} f = \sin^{1-j}(\alpha^{d-j+1})\partial_{\alpha^{d-j+1}}\big(\sin^{j-1}(\alpha^{d-j+1})\partial_{\alpha^{d-j+1}} f\big) + \csc^2(\alpha^{d-j+1})\Delta_{\mathbb{S}^{j-1}} f. \tag{4.20}$$

Here we have used the recurrence relation of the submatrix determinants of equation (4.17b). To obtain the eigenfunctions of the LAPLACE equation, we seek solutions to the HELMHOLTZ equation

$$\Delta_{\mathbb{E}^{d+1}}\Phi = -\kappa^2\Phi, \qquad \Phi \in L^2(\mathbb{E}^{d+1}). \tag{4.21a, b}$$

From equation (4.19), we find through multiplying by r^2 the confinement of κ to the radial part of the equation, perhaps lending some credence to an eigenvalue problem in the angular part separately. Indeed, since the unit hypersphere is compact, the angular part of the equation has a discrete spectrum. According to the spectral theorem,[2] the resolvent operator $(\mathrm{id} - \Delta_{\mathbb{S}^d})^{-1}$ is compact and selfadjoint,[3] with a countable spectral decomposition and the same eigenspace as the angular part of the hyperspherical HODGE–LAPLACE operator, yielding orthonormal basis with which to expand the operator in. That is, the full HODGE–LAPLACE operator preserves the eigenspace of the angular part. We may then construct the eigenspace iteratively by considering ever higherdimensional unit hyperspheres on which we solve the eigenvalue problem. For α^0, the operator is simply the second derivative, and the eigenvalue problem is the simple harmonic oscillator,

$$\partial^2_{\alpha^0}\Upsilon_{m_0} = -(m_0)^2\Upsilon_{m_0}, \qquad \Upsilon_{m_0}(\alpha^0) = \exp(im_0\alpha^0). \tag{4.22}$$

By the construction of the hyperspherical coördinates, in the sense that $\mathbb{S}^1 \subset \cdots \subset \mathbb{S}^d$, the ever hyperior hyperspherical harmonics, so to speak, are eigenfunctions for the lowerdimensional spaces as well. For higherdimensional operators, we write $\Upsilon^{m_{j-2},\ldots,m_0}_{m_{j-1}}$. By equation (4.20), we get the following STURM–LIOUVILLE problem for $j = 2$ when substituting $x = \cos\alpha^{d-1}$.

$$(1-x^2)F'' - 2xF' + \left(\lambda - \frac{(m_0)^2}{1-x^2}\right)F = 0, \tag{4.23}$$

where $\Upsilon^{m_0}_{m_1} = F\Upsilon_{m_0}$. This is the associated LEGENDRE equation, obtained by a substitution in the hypergeometric equation (4.6). We force regular solutions at $x = \pm 1$, yielding the eigenvalues $\lambda = m_1(m_1+1)$, such that $m_1 = |m_0| + k$ for all nonegative integers k. In general, we have that the angular hyperspherical eigenvalue problem through equation (4.20) takes the form of a GEGENBAUER equation, with solutions in the GEGENBAUER polynomials times a sinusoidal function.[4] It is customary to favor simple idices over the index chain for the hyperspherical harmonics, so that the upper index is m, and the lower index is n. We seek solutions in threedimensional Euclidean space, meaning we want the spherical harmonics on the two-sphere, which are[5,6]

$$\Upsilon^m_n(\alpha^0, \alpha^1) = c^m_n \,\mathrm{P}^{|m|}_n(\cos\alpha^1)\exp(im\alpha^0), \qquad c^m_n = (-1)^m\sqrt{\frac{2n+1}{4\pi}\frac{(n-|m|)!}{(n+|m|)!}} \tag{4.24a, b}$$

These constitute a complete orthonormal basis when $n \geq |m|$.[7]

4.4 Translation of harmonics

The matter of the translation and rotation of the spherical harmonic functions is treated in its entirety by GUMEROV & DURAISWAMI in a plethora of articles on the topic of multipole expansions and their swift evaluation by the fast multipole method. Chief among them is the paper wherein these rotations and translations are derived in detail,[8] their book on the fast multipole method applied to the HELMHOLTZ equation in particular serving as a readable alternative.[9] We shall however address first the matter of the

[1] [14] CHAVEL, pp.33–36
[2] [32] GRIGOR'JAN, thm.10.13; thm.10.20
[3] [19] CONWAY, prop.3.9; prop.4.13; thm.5.1
[4] [5] AVERY & AVERY, §3.5
[5] [16] CHENG, GREENGARD, & ROHLIN (1999), eq.4
[6] [36] GUMEROV & DURAISWAMI (2003), eq.2.5
[7] [75] TAKEUCHI, p.223
[8] [36] GUMEROV & DURAISWAMI (2003)
[9] [35] GUMEROV & DURAISWAMI, §3

result of the preceding text. We shall insert into equation (4.2) the relation (4.7) for the threedimensional case, yielding

$$\mathrm{G}(\boldsymbol{x};\boldsymbol{\varkappa}) = \frac{1}{4\pi|\boldsymbol{\varkappa}|}\sum_{k=0}^{\infty}\varepsilon^k\,\mathrm{P}_k(\cos\alpha). \tag{4.25}$$

We shall exploit the fact that the GEGENBAUER functions evaluated at the inner product between two higherdimensional vectors can be written as a superposition of products of the spherical harmonics.[1] This result simplifies further to what the literature on fast multipole calculations often labels the addition theorem for three dimensions, and is given as follows.

Theorem 4.1. *Let* P_n *denote the* LEGENDRE *polynomial of order* n, Υ_n^m *be a spherical harmonic, and* $\boldsymbol{x}_1$, $\boldsymbol{x}_2$ *be points on the unit two-sphere, such that* $\boldsymbol{x}_i = (\cos\alpha_i^1, \sin\alpha_i^0\sin\alpha_i^1, \cos\alpha_i^0\sin\alpha_i^1)$. *Then,*

$$\mathrm{P}_n(\cos\alpha) = \frac{4\pi}{2n+1}\sum_{m=-n}^{n}\Upsilon_n^m(\alpha_1^0,\alpha_1^1)\Upsilon_n^{m\dagger}(\alpha_2^0,\alpha_2^1), \tag{4.26}$$

where α *is the angle between the points.*

The coëfficient is due to the normalization condition imposed upon the spherical harmonics.[2] Note also the conjugate property that $\Upsilon^{m\dagger} = \Upsilon^{-m}$. We may now address the translation directly. It shall promtply become apparent that we seek local and distal expansions of the GREEN function. Relative to some new origin O' with vector $\boldsymbol{o}$ from the absolute origin, we write the point of evaluation $\boldsymbol{\varkappa}' = \boldsymbol{\varkappa} - \boldsymbol{o}$, the same with some source point $\boldsymbol{x}'$. The local and distal expansions shall respectively regular and singular at this new origin. In particular, we shall have that[3]

$$\mathrm{G}(\boldsymbol{x};\boldsymbol{\varkappa}) = \frac{1}{4\pi}\sum_{n=0}^{\infty}\sum_{m=-n}^{n}\mathrm{R}_n^m(-\boldsymbol{x}')\,\mathrm{S}_n^m(\boldsymbol{\varkappa}'), \qquad \varepsilon' < 1; \tag{4.27}$$

$$\mathrm{R}_n^m(\boldsymbol{\varkappa}') = \frac{(-1)^n i^{|m|}|\boldsymbol{\varkappa}'|^n}{c(n+|m|)!}\Upsilon_n^m(\boldsymbol{\varkappa}'), \qquad \mathrm{S}_n^m(\boldsymbol{\varkappa}') = \frac{(n-|m|)!}{i^{|m|}c|\boldsymbol{\varkappa}'|^{n+1}}\Upsilon_n^m(\boldsymbol{\varkappa}'). \tag{4.28a, b}$$

The arguments of the spherical harmonics is understood to be the angular part. Note the interpretation now of switching the arguments of the GREEN function—we are explicitly changing which of the two variables contribute regularly, and *vice versa* for thew singularity. The act of translating either harmonic by the regular and singular functions is to be interpreted as the reëxpansion of that harmonic locally or distally. In more concrete terms, we consider the position $\boldsymbol{o}_i$ of some origin one wishes to translate from, and the position $\boldsymbol{o}_j$ of some new origin one wishes to translate to. The displacement thereof we denote $\boldsymbol{d} = \boldsymbol{o}_j - \boldsymbol{o}_i$, such that the translations are given as follows.[4]

$$\mathrm{R}_n^m(\boldsymbol{d}+\boldsymbol{\varkappa}_j') = \sum_{n'=0}^{\infty}\sum_{m'=-n'}^{n'}\mathrm{R}_{n-n'}^{m-m'}(\boldsymbol{d})\,\mathrm{R}_{n'}^{m'}(\boldsymbol{\varkappa}_j') \tag{4.29}$$

$$\mathrm{S}_n^m(\boldsymbol{d}+\boldsymbol{\varkappa}_j') = \sum_{n'=0}^{\infty}\sum_{m'=-n'}^{n'}\mathrm{S}_{n+n'}^{m-m'}(\boldsymbol{d})\,\mathrm{R}_{n'}^{m'}(\boldsymbol{\varkappa}_j') \tag{4.30}$$

$$\mathrm{S}_n^m(\boldsymbol{d}+\boldsymbol{\varkappa}_j') = \sum_{n'=0}^{\infty}\sum_{m'=-n'}^{n'}\mathrm{R}_{n'-n}^{m-m'}(\boldsymbol{d})\,\mathrm{S}_{n'}^{m'}(\boldsymbol{\varkappa}_j') \tag{4.31}$$

For the translation of a singular function to one of regularity as with equation (4.30), we must evidently have that $|\boldsymbol{\varkappa}_j'| < |\boldsymbol{d}|$, and conversely for the translation of a singular expansion to another singular expansion as in equation (4.31), requiring that $|\boldsymbol{\varkappa}_j'| > |\boldsymbol{d}|$. Recall the magnitudinal ordering of the indices denoting the azimuthal FOURIER mode m, and the angular degree n. The multiplication of this azimuthal term in the spherical harmonic must yield the proper azimuthal index, whence the subtraction of the indices in the intermediary expansion about $\boldsymbol{d}$. The index in angular degree is similar in that regard, as the total degree of polynomials must additively coïncide due to the additive property of exponents. We shall later discuss these translations in terms of the translations needed for the fast multipole method, in the context thereof the customarily named fast multiple kernels of equations (4.29)–(4.31) are bestowed the following notation.

$$[\,\mathrm{R}\,|\,\mathrm{R}\,]_{n'n}^{m'm} = \mathrm{R}_{n-n'}^{m-m'}, \qquad [\,\mathrm{S}\,|\,\mathrm{R}\,]_{n'n}^{m'm} = \mathrm{S}_{n+n'}^{m-m'}, \qquad [\,\mathrm{S}\,|\,\mathrm{S}\,]_{n'n}^{m'm} = \mathrm{R}_{n'-n}^{m-m'} \tag{4.32a, b, c}$$

[1][5] AVERY & AVERY, §3.2
[2][23] EPTON & DEMBART (1995), eq.2.19
[3][33] GUMEROV & DURAISWAMI (2013), eq.25,28
[4][35] GUMEROV & DURAISWAMI, ch.5

We shall briefly also discuss an alternate, more geometrically inspired mode of translation, namely the method rotation, coäxial translation, and backrotation, or RCR for brevity. For a truncated multipole expansion, we let not n range to infinity, rather let its greatest extent be $N-1$. For some multipole coëfficient, we arrange for each value of n its span in m in an array. Consider now $\boldsymbol{d}$ given explicitly in terms of spherical coördinates, and the multipole coëfficient be given by A. Say we wish to translate from the singular expansion to an expansion in terms of the regular harmonic, and that this translation matrix T is given such that $A_{\mathrm{R}} = T_{\mathrm{S}|\mathrm{R}}(\boldsymbol{d})A_{\mathrm{S}}$. This translation matrix is composed of a rotation matrix R_{R}, a coäxial displacement matrix $C_{\mathrm{S}|\mathrm{R}}$, and the inverse rotation matrix R_{S}^{-1}. The matter of rotating the spherical harmonics is an involved, but not particularly tedious task. The rotation will preserve the degree of the spherical harmonic, but it will not preserve the order in general. That is, should we wish to rotate from the orientation (α^0, α^1) to the orientation (β^0, β^1), we have that

$$\Upsilon_n^k(\beta^0, \beta^1) = \sum_{m=-n}^{n} D_{mk}^n \Upsilon_n^m(\alpha^0, \alpha^1), \qquad D_{mk}^n = \exp(im\alpha^0) d_{mk}^n(\alpha^1) \tag{4.33}$$

Here, D_{mk}^n is the WIGNER function, and d_{mk}^n is the corresponding reduced function.[1] Thus for each value of n we have one submatrix of size $(2n+1)\times(2n+1)$, accounting for each of the $2n+1$ terms of m and their corresponding map to the same number of rotated terms. It is clear the rotation matrices will be diagonal, their coëfficient terms corresponding to regularization with respect to the spherical harmonics and either of the regular or singular functions. Along the diagonal, we then write

$$[R_{\mathrm{R}}^n]_k^m = \frac{i^{|k|}(n+|m|)!c_n^m}{i^{|m|}(n+|k|)!c_n^k} D_{mk}^n; \tag{4.34a}$$

$$[R_{\mathrm{S}}^n]_k^m = \frac{i^{|m|}(n-|k|)!c_n^m}{i^{|k|}(n-|m|)!c_n^k} D_{mk}^n. \tag{4.34b}$$

Since the translation $C_{\mathrm{S}|\mathrm{R}}$ is coäxial, we preserve the azimuthal structure of m, and we may thus entirely skip the offset in indexation in (4.32b). The final translation matrix is thus given in its decomposed form; $T_{\mathrm{S}|\mathrm{R}}(\boldsymbol{d}) = R_{\mathrm{R}}^{-1}(\alpha^0, \alpha^1) C_{\mathrm{S}|\mathrm{R}}(d) R_{\mathrm{S}}(\alpha^0, \alpha^1)$. In addition to its mathematical elegance, the sparsity of the rotation matrix, and the reduced cost in calculating the coäxial translation should theoretically reduce the needed computational strain from $O(N^4)$ to $O(N^3)$, yielding better performance with proper implementation.

5 Boundary element method

The boundary element method is a family of numerical methods wherein one discretizes the boundary upon which a calculation is to be made. For our intents and purposes, this entails calculating values related to the fluid velocity through the FREDHOLM integral equation of the second kind, equation (2.19) in particular. As has been established, the conditions under which the solution is unique is determined at the boundary, here being the NEUMANN boundary condition of impermeability,[2]

$$\Phi(\boldsymbol{x}) = \boldsymbol{U}\cdot\boldsymbol{x} + \Phi', \qquad \star\mathrm{d}\Phi = 0, \qquad \star\mathrm{d}\Phi' = -\star\mathfrak{U}, \tag{5.1a, b, c}$$

where $\boldsymbol{U}$ is the freestream velocity of the fluid, and $\mathfrak{U}$ is its corresponding one-form. Pulling back through $\iota_{\mathrm{b}}\colon S_{\mathrm{b}} \hookrightarrow \Omega$, where the boundary S_{b} is the LIPSCHITZ boundary of the impermeable body, is where we get equality in (5.1a, b). The orientation is induced by the outwardly pointing normal vector. We may then solve for the disturbance potential function Φ' through equation (2.16) by the means we see fit. For any square integrable vector field, we seldom choose any other methods than the integral equation representation of equation (2.19).[3,4,5] Many do insist, however, on the representation of the potential through a distribution of simple poles in the boundary, much in the same manner of our earlier treatment of the vorticity. This distribtuion is integrated against the GREEN function over the boundary to produce the value of the potential at some evaluation point, yielding an integral equation for the density through equation (2.19). The advantage of this approach to the problem of aerodynamic interaction between a body and the fluid is the automatic satisfaction of the LAPLACE equation for the disturbance away from the surface, and its evanescence. The source density is then a mechanism through which we enforce the

[1] [79] VARŠALOVIČ, MOSKALEV, & HERSONSKIJ, §4
[2] [8] BELIŠEV & ŠARAFUTDINOV (2008), §3
[3] [12] BYHOVSKIJ & SMIRNOV (1960)
[4] [63] MORINO (1993)
[5] [43] HESS & SMITH (1967)

impermeability condition, effectively acting as the instigator for fluid streamline displacement. Discretizing the body into qudrilateral panels, one then obtains algebraic equations, the numerical treatments of which consists of solving some integral through quadrature, and inverting an influence coëfficient matrix. Early treatments, pioneered in large part by Hess and Smith,[1] insisted on normal vectors coïnciding with their equivalent placements on the actual body before discretization. Thus, illfitting panels with incongruent edge nodes were sources of numerical errors.

The actual displacement of the fluid due to the existence of the wind turbine is not something of interest to us. In fact, we shall altogether ignore even the crassitude of the wind turbine blades, employing to some extent the thin wing approximation, and considering only the lifting surface that is the discretized mean camber line. The lifting surface we then treat as a cut on the manifold, now being the boundary. Doubtless, the potential cannot be expected to be continuous across the boundary here. Pulling back the potential from the boundary on either side should then reveal a scalar difference, which we shall label q_b. This jump in potential is merely in the nature of the dipole that is the integral of the exterior derivative of the Green function over the opposite sides of the lifting surface.

5.1 The vortex lattice method

The vortex lattice method was in name introduced by Falkner in a series of papers during the tailend of the 1940s.[2,3,4] This formulation, however, is entirely unsuited for unsteady vortex systems, as he therein postulates the vortex system to assume the form of Prandtl's steady horseshoe vortex system. It is formulated as follows. Consider a wing, be it swept or not, and section it spanwise with respect to the unswept axis into segments of ideally equal width. Place so, according to constant proportion of the chord, lines from root to tip of the wing at one quarter and three quarters of the chord, measured from leading edge to trailing edge, orthogonally to the segmentation of the wing. Draw then a vortex filament between the edges of the segments, through the quarterchord line for each segment. Emanating therefrom, along the edges of the segments towards the trailing edge and beyond to infinity, are vortex filaments. At infinity, we consider filaments meeting each of these emanating filaments, forming a closed vortex loop for each segment. The collection of these loops is what is known as the vortex lattice. At the threequarterschord, one will find the induced velocity of the vortex system, enforcing impermeability. For illustrative purposes, the unimaginative reader is referred to figure 1 in Falkner's 1947 report.

The reader would be remiss to not consider this business of extending the vortex filaments along the edges to infinity initially suspect in a discussion on unsteady aerodynamic schemes. For it is indeed the case that this initial formulation of the vortex lattice method is inherently steady. We have already discussed the fact that the inviscid formulation of the interaction between fluid flow and some solid object cannot account for the generation of vortical motion. As an *ad hoc* heuristic, the matter is often explained away by the prior shedding of a vortex, now carried downstream to infinity. Thus by the circulation theorem 2.19, the vorticity in the vicinity of the solid object must be of equal but negative magnitude, so that the total circulation of the system remains constant. Whence the vortex filament at infinity. Confer with the explanation of Newman.[5]

This peculiar placement of the bound vortex filament and control point for the evaluation of the downwash was first introduced in an article by Pistolesi,[6] which is unfortunately completely unavailable to the author. These placements were later shown to be optimal, even for wings of multiple chordwise panels, the vortex being placed at the quarterpoint, and the collocation point at the threequarterspoint.[7,8]

An albeit naïve yet illustrative explanation of the modern vortex lattice method is the exchange of the doublets of the standard boundary element method for vortex filaments along their boundaries. The mathematical foundation for this apparent equivalence between a dipole panel and a vortex panel is elusive in the literature. This relationship was to the author's knowledge first demonstrated by Hess,[9] Morino proving the very same in a later work,[10] relating the tangential derivative of the jump discontinuity in the potential over the boundary q to vorticity on the boundary. The intrinsic object on such a surface is indeed the exterior derivative $\mathrm{d}q$, its Hodge dual accordingly aligning with the previously discussed notion of vortex sheets. When the wake and bound panels have piecewise constant values for the potential jump, this exterior derivative vanishes in the calssical sense. The weak derivative, however, admits us to study the relationship between this potential jump and data at the edges of the panel through the

[1] [43] Hess & Smith (1967)
[2] [25] Falkner (1946)
[3] [27] Falkner (1947)
[4] [26] Falkner (1949)
[5] [65] Newman, §5.1, pp.169–172
[6] [22] DeYoung (1976)
[7] [73] Scholz (1950)
[8] [13] Byrd (1951)
[9] [41] Hess (1972), Appendix A
[10] [63] Morino (1993), Appendix C

fundamental theorem of exterior calculus.[1] Admittedly, the jumps in potential are treated in greater detail through the language of currents in the framework finite element exterior calculus,[2,3] which we shall lend credence to, but not discuss in any meaningful detail. Nevertheless, the weak derivative exists in a weak sense on the boundary skeleton. For the wake, we test the HODGE dual of the panelwise constant jump in potential against a smooth one-form, yielding

$$(\mathfrak{a}, \star \mathrm{d}q)_{S_\mathrm{w}} \equiv -\sum_{\mathrm{p}} q_\mathrm{p} \int_{\mathrm{p}} \mathrm{d}\mathfrak{a} = -\sum_{\mathrm{p}} q_\mathrm{p} \int_{\partial \mathrm{p}} \mathfrak{a} = \sum_{\mathrm{e}} \Gamma_\mathrm{e} \int_{C_\mathrm{e}} \mathfrak{a}(\boldsymbol{t}_\mathrm{e})\, \mathrm{d}s. \tag{5.2}$$

We have here used the fundamental theorem of multivariate calculus, and our preliminary work on vortical fluid flows, after first defining the test over the portion of the boundary that is the wake in a panelwise manner. Interior edges must thus be collected twice, with opposite orientations, yielding a difference on the edge that is exactly equivalent to the edge circulation. Edges whose neighboring panel does not exist, clearly yields no difference in potential jump from the panel value. In explicit mathematical terms, we have that for two adjacent panels p_1 and p_2, sharing the edge e between them, their respective values for the jump in potential over their interface being q_{p_1} and q_{p_2}, we define up to orientation the circulation on the shared edge $\Gamma_\mathrm{e} = q_{\mathrm{p}_2} - q_{\mathrm{p}_1}$, in a weak sense. Henceforth, we shall consider the induced velocity from a single panel

$$\boldsymbol{u}(\boldsymbol{\varkappa}) \approx \frac{1}{4\pi} \sum_{i=1}^{4} \Gamma_i \int_{C_i} \frac{\boldsymbol{x} - \boldsymbol{\varkappa}}{r^3} \times \mathrm{d}\boldsymbol{s},$$

where the circulation $\Gamma_i = q_\mathrm{p}$. For each filament, it is possible calculate the induced velocity analytically. In order to reduce clutter, we drop the index indicating filament. The procedure of evaluating for one filament in a lattice is illustrated in figure 5.

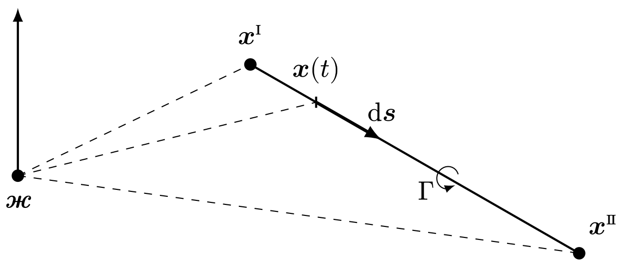


Figure 5: The calculation of the induced velocity from the vortex filament parametrized by $\boldsymbol{x}(s)$.

We consider the parametrization of the vortex filament with respect to the midpoint $\boldsymbol{\mathfrak{x}}$, with endpoints $\boldsymbol{x}^\mathrm{I}$ and $\boldsymbol{x}^\mathrm{II}$.[4] In contrast to the earlier treatment of parametrization of filaments, the arclength shall not serve as the parameter, us rather choosing $t \in [-1, 1]$. Since the filament is a straight line, we have that $\boldsymbol{x} = \boldsymbol{\mathfrak{x}} + {}^t\!/\!_2(\boldsymbol{x}^\mathrm{I} - \boldsymbol{x}^\mathrm{II})$. It can then be shown that the velocity induced by a straight vortex filament is given by

$$\boldsymbol{u}(\boldsymbol{\varkappa}) = \frac{\Gamma}{4\pi} \frac{\boldsymbol{r}^\mathrm{I} \times \boldsymbol{r}^\mathrm{II}}{r^\mathrm{I} r^\mathrm{II} + \boldsymbol{r}^\mathrm{I} \cdot \boldsymbol{r}^\mathrm{II}} \left(\frac{1}{r^\mathrm{I}} + \frac{1}{r^\mathrm{II}} \right), \tag{5.3}$$

where $\boldsymbol{r}^\mathrm{I} = \boldsymbol{\varkappa} - \boldsymbol{x}^\mathrm{I}$, and so on. Thus, the induced velocity at any point not on the filament can be calculated analytically. As the point of evaluation approaches the vortex filament, the scalar product in equation (5.3) approaches $-r^\mathrm{I} r^\mathrm{II}$, yielding infinite induced velocity, corresponding to the BIOT–SAVART kernel being singular. A remedy for the implementation for such an expression is to add a small correction proportional to the difference in the denominator,[5] or implementing a smoothing kernel for the vorticity two-form.[6] Opting for the former, one will write

$$\boldsymbol{u}(\boldsymbol{\varkappa}) = \frac{\Gamma}{4\pi} \frac{(\boldsymbol{r}^\mathrm{I} \times \boldsymbol{r}^\mathrm{II})(r^\mathrm{I} + r^\mathrm{II})}{r^\mathrm{I} r^\mathrm{II}(r^\mathrm{I} r^\mathrm{II} + \boldsymbol{r}^\mathrm{I} \cdot \boldsymbol{r}^\mathrm{II}) + (\varepsilon |\boldsymbol{r}^\mathrm{II} - \boldsymbol{r}^\mathrm{I}|)^2}, \tag{5.4}$$

where ε is the smoothing parameter. Figure 6 illustrates the impact of the smoothing parameter, both in terms of its magnitude, and the distance of the point of evaluation from some vortex filament. Ascending

[1] [11] BONALDI *et al.* (2025)
[2] [55] LICHT (2017)
[3] [15] CHEN, HUANG, & ZHANG (2025)
[4] [33] GUMEROV & DURAISWAMI (2013), Appendix B.1
[5] [47] KEBBIE-ANTHONY *et al.* (2019), §II.C
[6] [80] VATISTAS, KOZEL & MIH (1991)

from some isoline on the horizontal axis, it is evident that placing the point of evaluation close the the filament will increase the relative error in the induced velocity. In the same manner, increasing the smoothing parameter, perhaps better labeled the oft-called cutoff radius, increases the relative error overall.

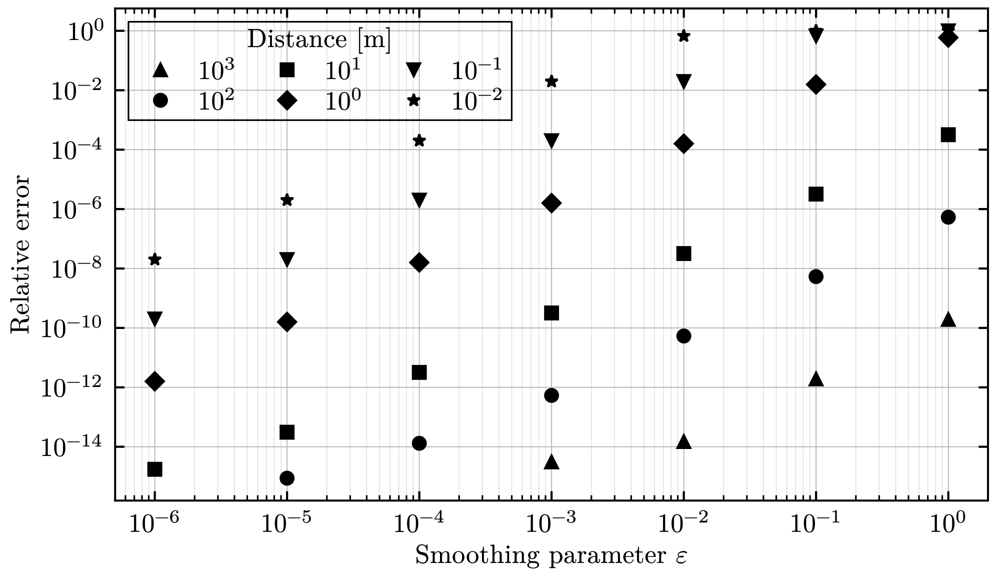


Figure 6: The relative error in induced velocity for different smoothing parameters, evaluated at varying distances from some vortex filament in a wind turbine wake.

Considering once more the mollified vorticity of equation (2.25), we can investigate a mollifier which trades the compact support for regularity at the filament. A collection of such possible mollifying functions have been discussed extensively in the literature.[1,2] We shall employ the mollifier of Rosenhead and Moore, which is the first of such modifications to the kernel conceived,[3,4] the actual expression adapted from Winckelmans and Leonard.[5] It is imperative that we be aware of the dimensionality of the problem at hand. For the literature treats two inherently different approaches in vortex mechanics in close proximity—one will find threedimensional distributions and twodimensional distributions, and it is the latter we shall employ. Recall that the vorticity be a two-form supported along a filament, us considering the flux through the surface area of some vortex tube thereof. The Rosenhead–Moore mollifier is then

$$\varphi_\sigma(n^1, n^2) = \frac{\sigma^2}{\pi(|n|^2+\sigma^2)^2}, \qquad \int_{\mathbb{R}^2} \frac{\sigma^2}{\pi(|n|^2+\sigma^2)^2}\,\mathrm{d}n^1\mathrm{d}n^2 = 1. \tag{5.5}$$

The velocity that results from adding this mollifier can be derived analytically,[6] and is what will be employed in the numerical machinery claculating the induced velocity from vortex lattices in the form of a wind turbine wake. It is clear that the length of the vortex filament is equal to the difference in length from the point of evaluation to either endpoint of the filament. That is, $\boldsymbol{s} = \boldsymbol{x}^{\mathrm{I}} - \boldsymbol{x}^{\mathrm{II}} = \boldsymbol{r}^{\mathrm{II}} - \boldsymbol{r}^{\mathrm{I}}$, with s denoting the magnitude of the vector, being the length of the filament. For clarity in the expression of the velocity, we then denote the distance from the evaluation point to the filament, and its distances to the endpoints as follows.

$$a = \frac{|\boldsymbol{r}^{\mathrm{I}} \times \boldsymbol{r}^{\mathrm{II}}|}{s}, \qquad b = \frac{\boldsymbol{r}^{\mathrm{I}} \cdot \boldsymbol{s}}{s}, \qquad c = \frac{\boldsymbol{r}^{\mathrm{II}} \cdot \boldsymbol{s}}{s} \tag{5.6}$$

The mollified velocity is then

$$\boldsymbol{u}_\sigma(\text{ж}) = \frac{\Gamma(\boldsymbol{r}^{\mathrm{I}} \times \boldsymbol{r}^{\mathrm{II}})}{4\pi s(\sigma^2+a^2)} \left(\frac{c}{\sqrt{\sigma^2+a^2+c^2}} - \frac{b}{\sqrt{\sigma^2+a^2+b^2}} \right). \tag{5.7}$$

[1] [53] Leonard (1985), eq.17
[2] [44] van Hoydonck, p.58
[3] [70] Rosenhead (1930)
[4] [62] Moore & Saffman (1972)
[5] [82] Winckelmans & Leonard (1993), eq.41; tab.II
[6] [44] van Hoydonck, Appendix F.1

5.2 Multipole expansions

The theoretical treatment of the LAPLACE equation in series predates the conceptual notion of much of the discussion of boundary element methods at hand. The application of equation (4.1), and the polynomials of LEGENDRE thereafter is an eighteenth century invention.[1] For our purposes, as we have stated yet not necessarily explained, the multipole expansion is simply an approximation tool with which we shall attempt to accelerate the calculation of farfield vortex filament influence on the local velocity. These are to be implemented through the local and distal spherical harmonic expansions R_n^m and S_n^m, respectively regular and singular at the point of evaluation. The matter of evaluating this velocity can be approached in two manners, one more natural to the standard multipole translation theory discussed earlier, and one more natural to the RCR translation. Both these methods exploit the HODGE decomposition of the velocity of equation (2.21a), the latter of which insists on the form (2.21b). In the literature of GUMEROV & DURAISWAMI, this particular latter form is referred to as the LAMB–HELMHOLTZ vector potential. Simply not applying the curl operator that results in equation (2.23) will evidently yield

$$\boldsymbol{\Psi}(\boldsymbol{ж}) = \Gamma \boldsymbol{t} I(\boldsymbol{ж}), \qquad I(\boldsymbol{ж}) = {}^{1}\!/\!{}_{2}|\boldsymbol{x}^{\mathrm{I}} - \boldsymbol{x}^{\mathrm{II}}| \int_{-1}^{1} \mathrm{G}\left(\boldsymbol{x}(t); \boldsymbol{ж}\right) \mathrm{d}t. \tag{5.8a, b}$$

The treatment of integrals like (5.8b) is a bottleneck for many implementations of the boundary element method, as the evaluation of integrals by quadrature tends to be rather ravenous in terms of its temporal extent, and lousy in its general accuracy. Exploiting the expansion of the GREEN function of equation (4.25), we may now expand this integral in terms of multipoles by the quadrature to expansion (Q2X) method,[2]

$$I(\boldsymbol{ж}) \approx \sum_{n=0}^{N-1} \sum_{m=-n}^{n} K_n^m(\boldsymbol{o})\, \mathrm{S}_n^m(\boldsymbol{ж}'), \qquad K_n^m(\boldsymbol{o}) = {}^{(-1)^n}\!/\!{}_{8\pi}|\boldsymbol{x}^{\mathrm{I}} - \boldsymbol{x}^{\mathrm{II}}| \int_{-1}^{1} \mathrm{R}_n^{-m}\left(\boldsymbol{x}'(t)\right) \mathrm{d}t. \tag{5.9a, b}$$

This expansion we know to be valid as long as the magnitude of the point of evaluation relative to some new origin is greater than the supremum of the magnitude of the point of expansion relative to that same new origin. Performing the actual evaluation of the integral (5.9b) is done through a recurrence relation.[3] The expansion coëfficients of the vector potential are then componentwise $\left[\Psi^i\right]_n^m = \Gamma t^i K_n^m$, such that the local vector potential after some translation is

$$\Psi^i(\boldsymbol{ж}) = \sum_{n=0}^{N-1} \sum_{m=-n}^{n} \left[\Psi^i\right]_n^m \mathrm{R}_n^m(\boldsymbol{ж}'). \tag{5.10}$$

This we shall henceforth label the *standard* velocity calculator, as it is the most obvious path from which we may calculate the velocity—we simply take the curl thereof. Due to the termwise nature of the differentiation of a series, we may perform the differentiation on either of the regular or singular functions, and transfer any index recurrence relation we presume we shall see to the multipole coëfficients by reïndexing. We shall reïntroduce the complex differential in terms of the variable $ȝ = ч + iз$ for the local frame $\boldsymbol{ж} = ж\,\partial_x + ч\,\partial_y + з\,\partial_z$, constructing a cylindrical coördinate system with respect to the complex variable. We then consider the generating function of the regular function, derived from the HERGLOTZ function,[4,5]

$$f(a, b; \boldsymbol{ж}) = \exp\left(a\left(-ж + \frac{ȝ}{2b} - \frac{bȝ^\dagger}{2}\right)\right) = \sum_{n=0}^{\infty} \sum_{m=-n}^{n} a^n b^m\, \mathrm{R}_n^m(\boldsymbol{ж}). \tag{5.11}$$

It is clear that differentiating the explicit exponential representation of the function f of equation (5.11) will result in a selfsimilar differential. Indeed, $\partial_ж f = -af$ and $\partial_ȝ f = {}^{a}\!/\!{}_{2b} f$ and $\partial_{ȝ^\dagger} f = -{}^{ab}\!/\!{}_{2} f$. Due to the uniform and absolute convergence of the series representation for nonzero b, we may differentiate the regular functions term by term, assigning recurrence relations based on the principle of homogeneity we have discussed earlier. That is, $\partial_ж \mathrm{R}_n^m = -\mathrm{R}_{n-1}^m$ and $\partial_ȝ \mathrm{R}_n^m = {}^{1}\!/\!{}_{2}\,\mathrm{R}_{n-1}^{m+1}$ and $\partial_{ȝ^\dagger} \mathrm{R}_n^m = -{}^{1}\!/\!{}_{2}\,\mathrm{R}_{n-1}^{m-1}$. We find after returning to Cartesian coördinates that

$$\mathrm{dR}_n^m = -\mathrm{R}_{n-1}^m\,\mathrm{d}x + \left(\frac{\mathrm{R}_{n-1}^{m+1} - \mathrm{R}_{n-1}^{m-1}}{2}\right)\mathrm{d}y - \left(\frac{\mathrm{R}_{n-1}^{m+1} + \mathrm{R}_{n-1}^{m-1}}{2i}\right)\mathrm{d}z \tag{5.12}$$

[1] [48] KELLOGG, pp.124–146
[2] [37] GUMEROV, KANEKO, & DURAISWAMI (2023)
[3] [37] GUMEROV, KANEKO, & DURAISWAMI (2023), eq.43
[4] [71] SACK (1974)
[5] [20] COURANT & HILBERT, §VII.7.3

It ought now to be evident that the calculation of the velocity from equation (5.10) is no more complicated than evaluating the curl thereof. We shall however but for a brief moment clarify the matter of differentiating a scalar function of expanded form. Let F be some scalar function for which we have some truncated local expansion of maximal degree $N-1$, relative to another origin, such that we may express it in terms of the regular function as follows.

$$F(\text{ж}) = \sum_{n=0}^{N-1} \sum_{m=-n}^{n} L_n^m \, \mathrm{R}_n^m(\text{ж}')$$

As was stated earlier, differentiating this function will result in a shift of indices, which we may inversely apply to the local multipole coëfficients such that they take the form

$$[\mathrm{d}F]_n^m = -L_{n+1}^m \,\mathrm{d}x + \left(\frac{L_{n+1}^{m-1} - L_{n+1}^{m+1}}{2}\right) \mathrm{d}y - \left(\frac{L_{n+1}^{m-1} + L_{n+1}^{m+1}}{2i}\right) \mathrm{d}z \tag{5.13}$$

From the relations for the regular derivative of equation (5.12), it is trivial to find the differential recurrence relations for the singular function by considering the multipole expansion of the Green function in equation (4.25). Transferring the differential operator instead to the regular function, we in much the same manner as with the regular differential recurrence find the recurrence for the singular function, though of opposite index.

$$\mathrm{dS}_n^m = -\,\mathrm{S}_{n+1}^m \,\mathrm{d}x + \left(\frac{\mathrm{S}_{n+1}^{m+1} - \mathrm{S}_{n+1}^{m-1}}{2}\right) \mathrm{d}y - \left(\frac{\mathrm{S}_{n+1}^{m-1} + \mathrm{S}_{n+1}^{m+1}}{2i}\right) \mathrm{d}z \tag{5.14}$$

This formulation is in general less useful, as the derivative is something we would like to calculate as a final step after multipole translation. Nevertheless, expanding the scalar function F now distally, we get the multipole coëfficients M_n^m. In the exact same manner as with the local expansion, we shift the indices in the singular function, effectively moving the differentiation onto the multipole coëfficients. We then have

$$[\mathrm{d}F]_n^m = -M_{n-1}^m \,\mathrm{d}x + \left(\frac{M_{n-1}^{m-1} - M_{n-1}^{m+1}}{2}\right) \mathrm{d}y - \left(\frac{M_{n-1}^{m-1} + M_{n-1}^{m+1}}{2i}\right) \mathrm{d}z. \tag{5.15}$$

We may now find the standard velocity by taking the curl of the vector potential, calculating the differentials componentwise according to the formula previously discussed. We write the velocity one-form $\mathfrak{u} = u\,\mathrm{d}x + v\,\mathrm{d}y + w\,\mathrm{d}z$, such that we may calculate the multipole coëfficients of these components as follows.

$$u_n^m = [\partial_y \Psi^z]_n^m - [\partial_z \Psi^y]_n^m = \frac{[\Psi^z]_{n+1}^{m-1} - [\Psi^z]_{n+1}^{m+1}}{2} + \frac{[\Psi^y]_{n+1}^{m-1} + [\Psi^y]_{n+1}^{m+1}}{2i} \tag{5.16a}$$

$$v_n^m = [\partial_z \Psi^x]_n^m - [\partial_z \Psi^z]_n^m = [\Psi^z]_{n+1}^m - \frac{[\Psi^x]_{n+1}^{m-1} + [\Psi^x]_{n+1}^{m+1}}{2i} \tag{5.16b}$$

$$w_n^m = [\partial_x \Psi^y]_n^m - [\partial_y \Psi^x]_n^m = \frac{[\Psi^x]_{n+1}^{m+1} - [\Psi^x]_{n+1}^{m-1}}{2} - [\Psi^y]_{n+1}^m \tag{5.16c}$$

We have established the Hodge deocmposition theorem as a matter of fact insofar as we may indeed express the velocity one-form as the sum of the exterior derivative of some scalar function, the codifferential of some one-form, and some harmonic one-form. The expression for the velocity just derived is a choice of gauge in the velocity vector potential Ψ, the form of which we have made purposefully unspecified. Forcing the specificity of the vector potential will yield some corrective exact form of sorts, yielding the so-called Lamb–Helmholtz representation of the velocity, as we specified in equation (2.21).[1] The advantage of this gauge is the fact that we now only have to calculate two scalar potentials Φ and χ, compared to the three components of the vector potential $\boldsymbol{\Psi}$. We must then have that from any filament the velocity induced is

$$\mathfrak{u} = \star(\mathrm{d}I \wedge \Gamma\mathfrak{t}) = \mathrm{d}\Phi + \star\mathrm{d}(\chi \text{ж}'^{\flat}); \tag{5.17}$$

$$\Phi(\text{ж}) = \sum_{n=0}^{\infty} \sum_{m=-n}^{n} \Phi_n^m \, \mathrm{S}_n^m(\text{ж}'), \qquad \chi(\text{ж}) = \sum_{n=0}^{\infty} \sum_{m=-n}^{n} \chi_n^m \, \mathrm{S}_n^m(\text{ж}'). \tag{5.18a, b}$$

[1] [51] Lamb, §335, eq.9

We need now only contract either side of this equation with the vector $\boldsymbol{ж}'$ to isolate the scalar potential. Using the fact that $\boldsymbol{a} \lrcorner \star\mathfrak{b} = \star(\mathfrak{b} \wedge \mathfrak{a})$, we must indeed have that $\boldsymbol{ж}' \lrcorner \star(\mathrm{d}\chi \wedge \boldsymbol{ж}'^\flat) = 0$. Taking the curl evidently erases the scalar potential term. The other term we expand as $\bar{\mathrm{d}}\mathrm{d}(\chi \boldsymbol{ж}'^\flat) = \mathrm{d}(\chi + \boldsymbol{ж}' \lrcorner \mathrm{d}\chi) - \boldsymbol{ж}'^\flat \Delta\chi$. In the same vein, we write that $\bar{\mathrm{d}}(\mathrm{d}I \wedge \mathfrak{t}) = \mathrm{d}(\boldsymbol{t} \lrcorner \mathrm{d}I) - \mathfrak{t}\Delta I$. The integral I is clearly harmonic by construction, and the potential χ is harmonic by gauge. Equating primitives, we find that the procedure is

$$\star(\Gamma \mathrm{d}I \wedge \mathfrak{t} \wedge \boldsymbol{ж}'^\flat) = \boldsymbol{ж}' \lrcorner \mathrm{d}\Phi, \qquad \Gamma \boldsymbol{t} \lrcorner \mathrm{d}I = (\mathrm{id} + \boldsymbol{ж}' \lrcorner \mathrm{d})\chi. \tag{5.19a, b}$$

The singular function is homogeneous, meaning $\mathrm{S}_n^m(a\boldsymbol{ж}') = |a|^{-(n+1)}\,\mathrm{S}_n^m(\boldsymbol{ж}')$, and that the direcitonal derivative at $a = 1$ yields $\boldsymbol{ж}' \lrcorner \mathrm{d}\,\mathrm{S}_n^m = -(n+1)\,\mathrm{S}_n^m$. We may now use the relations of equation (5.14), and shift the indices as before when performing the derivative of the line integral. Thus, for the source terms of the potentials, we find their expansion coëfficients to be given in terms of the line integral multipole coëfficients of equation (5.9b) as follows.[1]

$$\Phi_n^m = \frac{\Gamma}{2(n+1)}\Big(2imt^x K_n^m + \big(-t^z + it^y\big)(n-m+1)K_n^{m-1} + \big(t^z + it^y\big)(n+m+1)K_n^{m+1}\Big) \tag{5.20}$$

$$\chi_n^m = -\frac{\Gamma}{2n}\Big(-2t^x K_{n-1}^m + t^y\big(K_{n-1}^{m-1} - K_{n-1}^{m+1}\big) + it^z\big(K_{n-1}^{m+1} + K_{n-1}^{m+1}\big)\Big) \tag{5.21}$$

For the evaluation of the curl in terms of the regular expansion, we again use the generating function of equation (5.11). We shall seek expressions of the form $\star(\mathrm{d}f \wedge \boldsymbol{ж}'^\flat)$, which we may simply calculate the differentials of the generating function. Indeed, we find that $ib\partial_b f = з\partial_ч f - ч\partial_з f$, which is exactly the component of the curl along $ж$. Matching coëfficients of the regular harmonic for this and the remaining components, we find that

$$u_n^m = -\Phi_{n+1}^m + im\chi_n^m; \tag{5.22a}$$

$$v_n^m = \frac{\Phi_{n+1}^{m-1} + \Phi_{n+1}^{m+1}}{2} + \frac{(n+m)\chi_n^{m-1} - i(n-m)\chi_n^{m+1}}{2i}; \tag{5.22b}$$

$$w_n^m = \frac{(n+m)\chi_n^{m-1} - (n-m)\chi_n^{m+1}}{2} - \frac{\Phi_{n+1}^{m-1} + \Phi_{n+1}^{m+1}}{2i}. \tag{5.22c}$$

A considerable weakness of this method of calculating the velocity components is its dependence on the vector $\boldsymbol{ж}$ in its gauge. Translating the functions will in general break the gauge, and so it must be repaired.[2] Let the old expansion origin be $\boldsymbol{o}_i$, and the new be $\boldsymbol{o}_j$, such that $\boldsymbol{d} = \boldsymbol{o}_j - \boldsymbol{o}_i$, as before, and the translated potentials be denoted $\hat{\Phi}$ and $\hat{\chi}$. Now we evidently have that $\boldsymbol{ж}_i' = \boldsymbol{d} + \boldsymbol{ж}_j'$, and that the potentials satisfy $\mathrm{d}(\Phi - \hat{\Phi}) + \star\mathrm{d}\big((\chi - \hat{\chi})\boldsymbol{ж}_j'^\flat\big) = \star\mathrm{d}(\hat{\chi}\boldsymbol{d}^\flat)$. Exploiting the same procedure as we did for the untranslated coëfficients, we find that we must have

$$\boldsymbol{ж}_j' \lrcorner \mathrm{d}(\Phi - \hat{\Phi}) = \boldsymbol{ж}_j' \lrcorner \star(\mathrm{d}\hat{\chi} \wedge \boldsymbol{d}^\flat), \qquad (\mathrm{id} + \boldsymbol{ж}_j' \lrcorner \mathrm{d})(\chi - \hat{\chi}) = \boldsymbol{d} \lrcorner \mathrm{d}\hat{\chi}. \tag{5.23a}$$

Gumerov & Duraiswami now introduce conversion coëfficients to convert the translated potentials back into Lamb–Helmholtz form. For the regular and singular harmonics we must then have corresponding conversion coëfficients, each tailored to the respective differential properties of the function. For brevity, we shall write $A_n^m = \big[\boldsymbol{ж}_j' \lrcorner \star(\mathrm{d}\hat{\chi} \wedge \boldsymbol{d}^\flat)\big]_n^m$, and $B_n^m = \big[\boldsymbol{d} \lrcorner \mathrm{d}\hat{\chi}\big]_n^m$. As we have already found, normal differentiation in the regular function yields a multiplicative factor of n, and so

$$\Phi_n^m = \hat{\Phi}_n^m + \frac{A_n^m}{n}, \qquad n \geq 1; \tag{5.24a}$$

$$\chi_n^m = \hat{\chi}_n^m + \frac{B_n^m}{n+1}, \qquad n \geq 0, \tag{5.24b}$$

where we set $\Phi_0^0 = \hat{\Phi}_0^0$. We copmute these explicitly as follows.

$$A_n^m = -imd^ж\hat{\chi}_n^m + \frac{(n+m)\hat{\chi}_n^{m-1}}{2}\big(id^ч - d^з\big) - \frac{(n-m)\hat{\chi}_n^{m+1}}{2}\big(id^ч + d^з\big) \tag{5.25a}$$

$$B_n^m = -d^ж\hat{\chi}_{n+1}^m + \frac{d^ч\big(\hat{\chi}_{n+1}^{m-1} - \hat{\chi}_{n+1}^{m+1}\big)}{2} - \frac{d^з\big(\hat{\chi}_{n+1}^{m-1} - \hat{\chi}_{n+1}^{m+1}\big)}{2i} \tag{5.25b}$$

[1][33] Gumerov & Duraiswami (2013), eq.47

[2][33] Gumerov & Duraiswami (2013)

For the singular harmonic expansion, we similarly have the following, though with the condition that $\chi_0^0 = \hat{\chi}_0^0$.

$$\Phi_n^m = \hat{\Phi}_n^m + \frac{A_n^m}{n+1}, \qquad n \geq 0 \tag{5.26a}$$

$$\chi_n^m = \hat{\chi}_n^m + \frac{B_n^m}{n}, \qquad n \geq 1 \tag{5.26b}$$

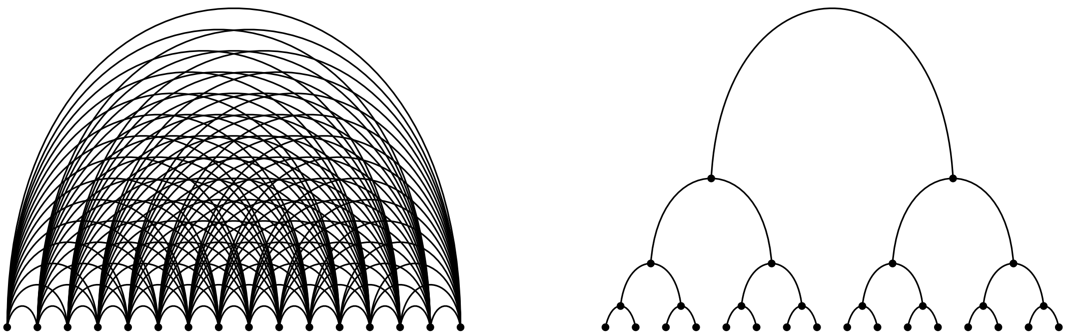
$$A_n^m = -imd^{\text{ж}}\hat{\chi}_n^m - \frac{(n-m+1)\hat{\chi}_n^{m-1}}{2}(id^{\text{ч}} + d^{\text{з}}) + \frac{(n+m+1)\hat{\chi}_n^{m+1}}{2}(id^{\text{ч}} - d^{\text{з}}) \tag{5.27a}$$

$$B_n^m = -d^{\text{ж}}\hat{\chi}_{n-1}^m + \frac{d^{\text{ч}}(\hat{\chi}_{n-1}^{m-1} - \hat{\chi}_{n-1}^{m+1})}{2} - \frac{d^{\text{з}}(\hat{\chi}_{n-1}^{m-1} - \hat{\chi}_{n-1}^{m+1})}{2i} \tag{5.27b}$$

We shall for the reimainder of this work refer to this construction of the velocity as the Lamb–Helmholtz velocity, and as was stated earlier, we refer to the somewhat more direct method with the three scalar potentials as the standard method. With now either of the velocity constructions, and the two translation methods, we have four options for the implementation of the fast multipole method at our disposal.

5.3 The fast multipole method

The fast multipole method was developed by Leslie Greengard and Vladimir Rohlin in their seminal paper *A Fast Algorithm for Particle Simulations*.[1] The application was manyparticle systems, interstellar mechanics and electrostatics being the immediate examples thereof. Consider a swath of electrically charged particles gathered in clusters at some locations, and spread war apart in others. By the law of Coloumb governing the physics of charged particle interaction, the force impressed upon some particle by all the others depends on its distance thereto. The idea of the fast multipole method is to systematically evaluate the influence of faraway sources on a point by aggregating their influence, and calculating them in bulk. This is done through octree space partitioning. Whatever influence an aggregate of points children nodes have are passed onto the parent, and the influence of the parent is calculated instead. This is efficient computationally because a lot of the required calculation for each panel is already done. For the unsteady vortex lattice method, all vertices of wake filaments are to be convected farther downstream, influenced by the entirety of the wake, meaning the total number of calculations needed for a single step in time is proportional to the square of the number of such vertices. This type of interaction, where the evaluation of each point depends on some property of every other point is illustrated in figure 7a.[2] By instead considering instead the tree-based evaluation of the fast multipole method through aggregation of neighboring influence as show in 7b, the complexity of the number of operations in the evaluation is now shifted over to the complexity of the mechanism by which one aggregates and translates the influence through the tree.



(a) Direct interactions.

(b) Tree-based interactions.

Figure 7: The difference in number of interactions for the direct and tree-based approaches.

[1] [31] Greengard & Rohlin (1987)

[2] [57] Liu, fig.3.1, p.49, adapted with permission from the author.

6 Numerical implementation

The numerical implementation that has been implemented has been somewhat tailored to process data output by `3DFloat` provided by the main advisor.[1] Since the very purpose of this work has in large part been to determine whether the fast multipole method is worth the effort it takes to implement in terms of numerical accuracy compared to the cost of computation, we are to compare the results of computation of the fast multipole accelerated vortex lattice method to the direct calculation thereof. That is to say that we are given a partial solution produced by `3DFloat`, for which we calculate the induced velocity directly, and quickly using the fast multipole method. Output from this partial run in `3DFloat` is a `vtk`-file with the vertices of the panels of the wake, along with their potential jump values and connectivity. The run was started abruptly, with time step sizes of 0.02 s, releasing panels into the wake at every twentieth step in time, the total elapsed time being 1500 s.

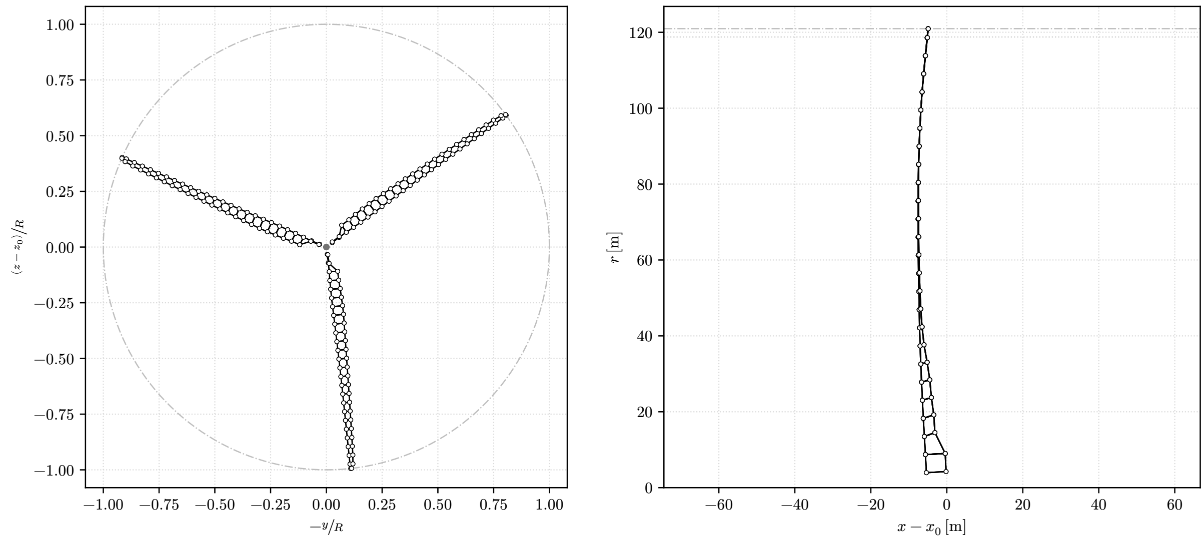


Figure 8: The turbine blades from the `vtk`-file.

The modeled wind turbine is an IEA 15 MW, with a rotor diameter about of $D = 234$ m, though the measured radius of the blades in the `vtk`-file yield a radius $R = 120.945$ m, which shall serve as the baseline. Extracting the data for the blades as we have in figure 8, we see that it has some twist near the root. As such we shall choose the domain of evaluation somewhat carefully. It is worth noting that the solid lines are inferred vortex filaments, and are not part of the data, for the "circulation" described at each panel of connected vertices is indeed merely the potential jump, and not the edge circulation we derived earlier. The run we are provided with has a nontrubulent freestream velocity $10\,\mathrm{m\,s^{-1}}$, being a representative simulation of wind speeds below rate.

What follows is a description of the methodology for the implementation of the fast multipole accelerated vortex lattice method, and the results of tests performed. We make some cloud of target points for which we shall calculate the induced velocity, and partition the domain in a manner which classifies wellseparated filaments as such. The potential induced by the filaments may then be expanded into multipole expansions, and be translated in such a manner that they may be evaluated at the target points.

6.1 Octree data structure

An *octree* is a data structure which partitions the cube, whose modern adaptation in large part can be attributed to a 1980 report by Donald Meagher.[2] Owing to its treelike structure and roots in other arborial algorithms, so to speak, the parlance with which one speaks of it is thusly inspired. The partitioning starts with a *root*, which is split into eight equal pieces, whence its name—we shall call these cubes *voxels*. Like a family, these voxels can produce further children voxels, their relation thereto being that of a parent. Imagining an inverted structure compared to that of a physical tree, we place the root voxel at the very top, and let its eight children spread out downwards. These voxels shall then perhaps

[1] [67] Nygaard emphet al. (2016)

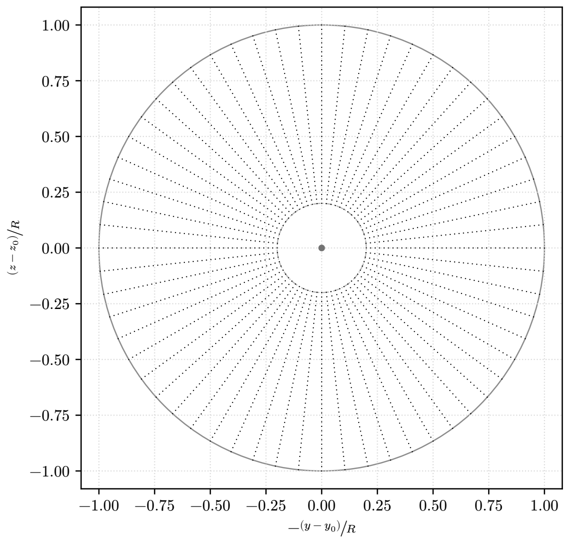
[2] [59] Meagher (1980)

also have eight children, and so on. Voxels that have no children we call leaves. For each generation, we may assign a depth relative to the root, whose depth is zero, and the maximal depth in the tree we call the *profundity*. Algorithm 1 outlines the implemented construction, adhering to a conservative subdivision criterion of voxel ownership of both endpoints of the filaments to insure regularity.

The purpose of implementing this partitioning is to effectively determine sufficient distance between targets and sources, and to aggregate the sources by means of multipole transfers. The direct application to the problem at hand in this thesis is to somehow partition the domain of evaluation using the octree data structure, and expanding the influence coëfficients of the filaments relative to the voxel center. Calculating the influence of some aggregate of filaments is then as simple as translating the coëfficients up the tree, and down again to the target. This is referred to as the upward and downward passes, respectively. Determining what constitutes farfield is an involved matter, which has been addressed through so-called `U`, `V`, `W`, and `X` lists.[1,2] The construction of these interaction lists is specified in algorithm 2. Further, we shall adopt the parlance of the fast multipole method, denoting the translations of equation (4.32) respectively the L2L, M2L, and M2M translations. These names are perhaps more explanatory than their purely mathematical counterparts. The M2M, or multipole-to-miltipole translation shifts the child multipoles to the center of the parent voxel, and so on up the octree, until the downward pass of the M2L, or multipole-to-local translation of this wellseparated source multipole to a local regular representation. The L2L, or local-to-local translation shifts the parent local expansion to the child target center.

6.2 Results

We construct a cylindrical coördinate system according to right-hand orientation. The x-axis will serve as the cylindrical extension of the polar yz-plane, centered at the hub of the wind turbine rotor (x_0, y_0, z_0), downstream of the fluid flow being in the direction of increasing x, as is illustrated in figure 8. For all tests we shall perform, we shall distribute points along the rotorplane in a linear manner along the radial axis for ${}^{r}/_{R} \in [{}^{1}/_{5}, 1]$, and along the azimuthal axis for $\alpha \in [0, 2\pi)$, where the ray of zero degrees coïncides with the vertical z-axis. The number of points in the radial direction we choose to correspond approximately to one point per meter of blade, yielding a total number of 121 pints, regardless of the span of the interval of evaluation. In the azimuthal direction, we place one point at every other degree, yielding a total of 180 points. The grand total of points is thus $121 \times 180 = 21,780$. A selection of these numerous points is shown in figure 9. It is worth noting that the total number of panels in the `vtk`-file is 268,650, yielding the number of filaments to be 1,074,600. These points are chosen according to a spline interpolation of the midpoint of the wind turbine blades, which is then sampled in the radial axis, and swept in the azimuthal direction to construct a faux rotorplane.



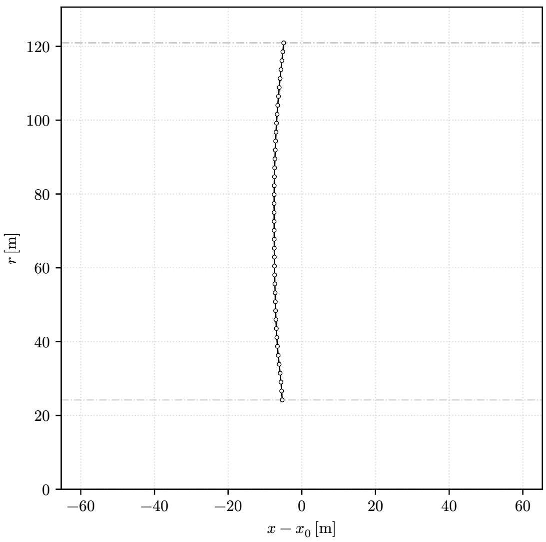


Figure 9: The evaluation rotorplane. Every third evaluation point is shown.

After the evlauation of the induced velocity in the rotorplane, placing it at some specified position in the wake, measured in ${}^{x}/_{D}$, the data is postprocessed in `json`-files containing the run data, and in

[1] [16] Cheng, Greengard, & Rohlin (1999)

[2] [81] Wala & Klöckner (2018)

npz-files containing the induced velocity. Representing the data graphically, we have chosen two methods of displaying the induced velocity. First is the induced velocity at $r/R = 0.8$, graphed over the azimuthal variable, illustrating the directly calculated values, with quickly calculated values of some chosen fast multipole method superimposed. Second is the average over the r axis, weigted according to the cylindrical Jacobian $r\,\mathrm{d}r\,\mathrm{d}\alpha$. We shall test four such placements along the wake, namely values of x corresponding to $-1D$, $-0.1D$, $0D$, and $1D$. That is, one rotor diameter upstream of the rotorplane, onetenth rotor diameters upstream from the rotorplane, the rotorplane itself, and one rotor diameter downstream of the rotorplane. We choose to compare the direct evaluation of the induced velocity to the LAMB–HELMHOLTZ representation of the velocity, translated with the RCR method, at multipole order eight. We shall also inspect the convergence rates of the different methods at each of these placements, measured with respect to the relative error in the velocity. This measure of error is the standard choice in the fast multipole literature, and is calculated by taking the ℓ^2 magnitude of the difference of the direct and quick induced velocities, divided by the ℓ^2 magnitude of the direct induced velocity.

For the case of one rotor diameter upstream of the rotorplane, we expect the induced velocity in the azimuthal direction to be effectively zero, due to the absence of viscosity. Inspecting the data of figures 10 and 11, we observe nothing unexpected. There is indeed no induced velocity in the azimuthal direction, only relatively small but expected velocities in the radial and transversal axes. The multipole evaluation is virtually indistinguishable, being sampled sparsely to be superimposed, so as to not visually interfere with the direct calculation. The relative error is quite good, but not excellent at order four, as seen in figure 12, and the minimal error seems to get no smaller than on the order of 10^{-8}.

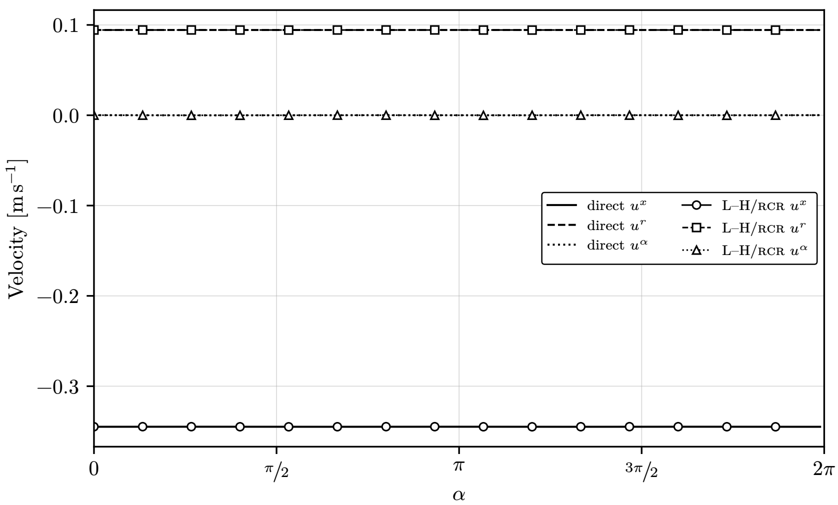


Figure 10: The induced velocity at $x/D = -1$, for $r/R = 0.8$.

At a value of x corresponding to $-0.1D$, still do not expect to see any azimuthal induced velocity, again due to the inviscid nature of the flow. It is clear from the plot in figures 13 and 14, however, that there is some induced velocity in the azimuthal direction. We now see the three times periodic behavior expected from the presence of three distinct convected wake surfaces. The convergence of the relative error of the multipole expansions is now improved approximately by an order of magnitude, as is clear from figure 15

At the rotorplane, there seem to be spikes in velocity around

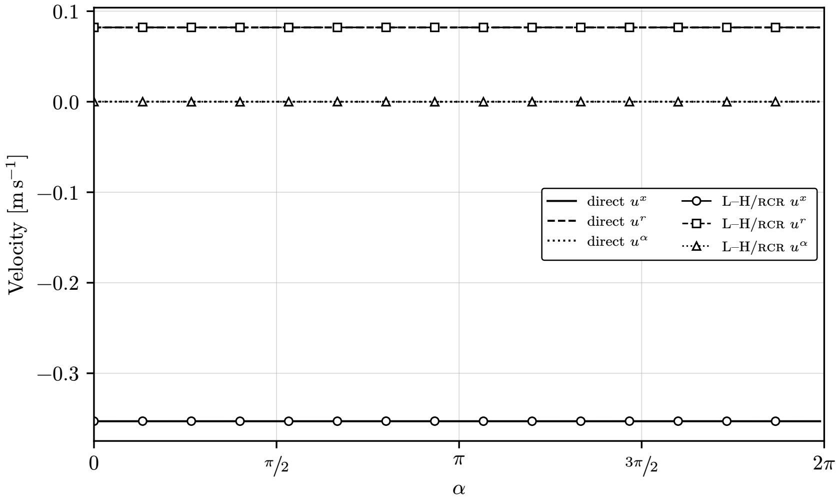


Figure 11: The induced velocity at $x/D = -1$, averaged radially.

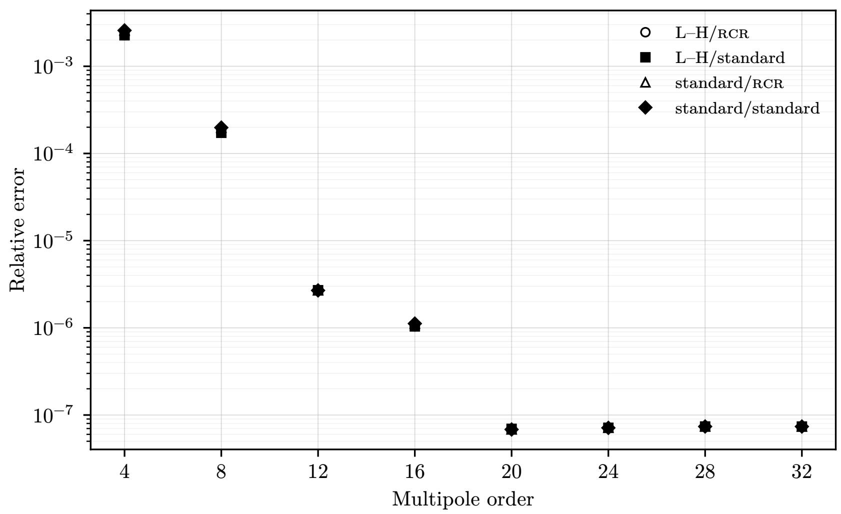


Figure 12: The relative error of the multipole induced velocity to the direct calculation at $x = -1D$.

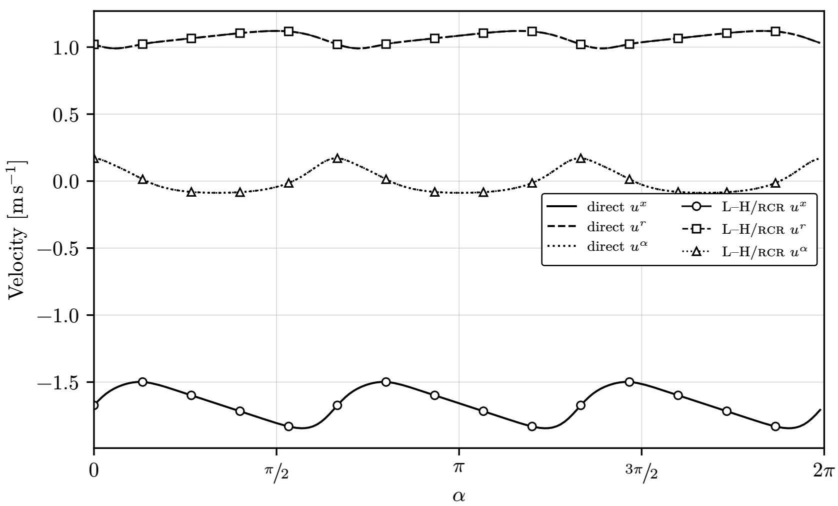


Figure 13: The induced velocity at $x/D = -0.1$, for $r/R = 0.8$.

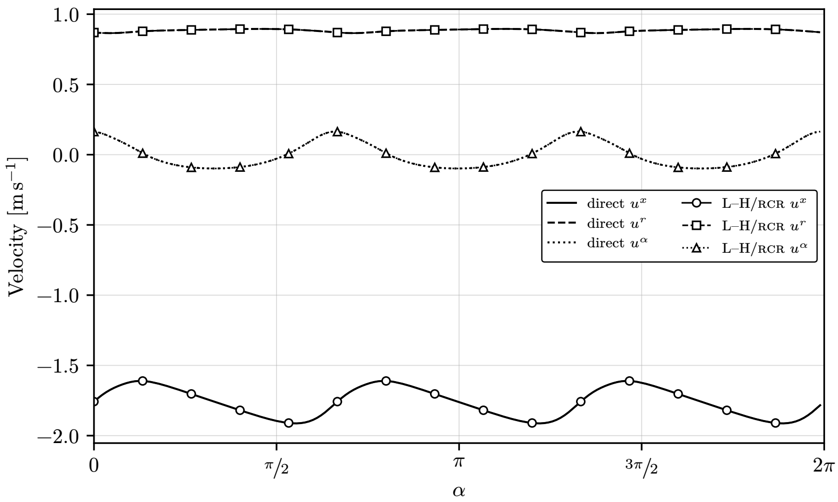


Figure 14: The induced velocity at $x/D = -0.1$, averaged radially.

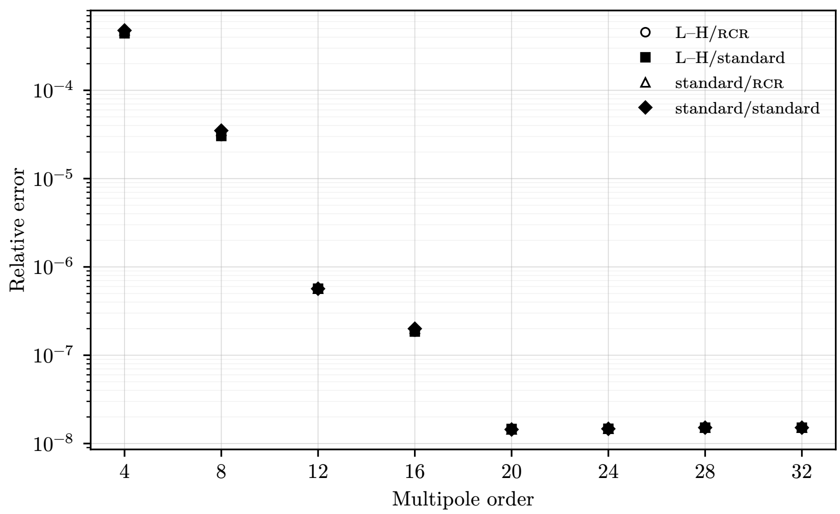


Figure 15: The relative error of the multipole induced velocity to the direct calculation at $x = -0.1D$.

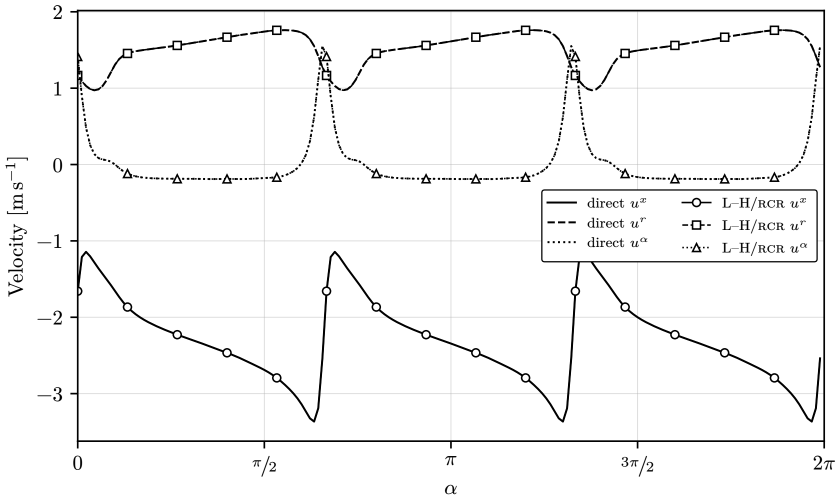


Figure 16: The induced velocity at the rotorplane, for $r/R = 0.8$.

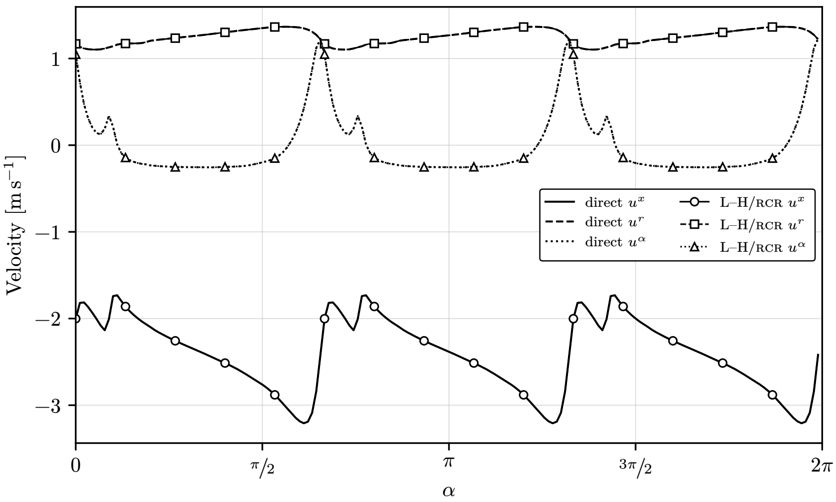


Figure 17: The induced velocity at the rotorplane, averaged radially.

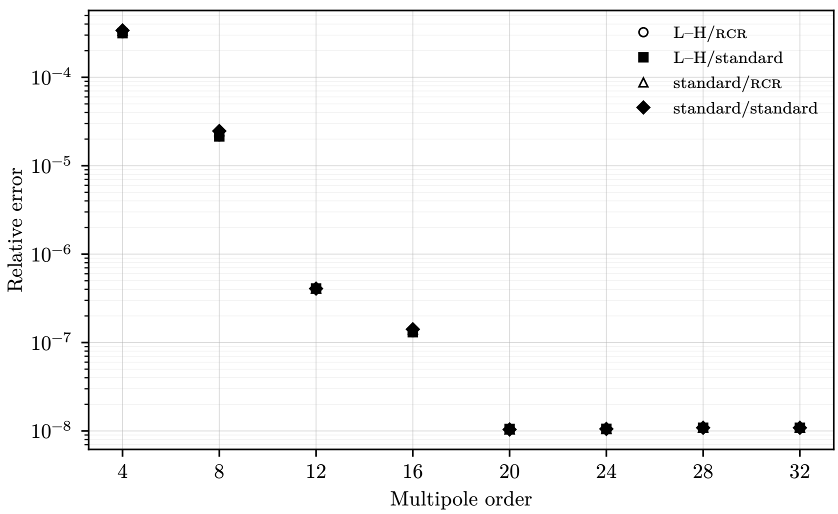


Figure 18: The relative error of the multipole induced velocity to the direct calculation in the rotorplane.

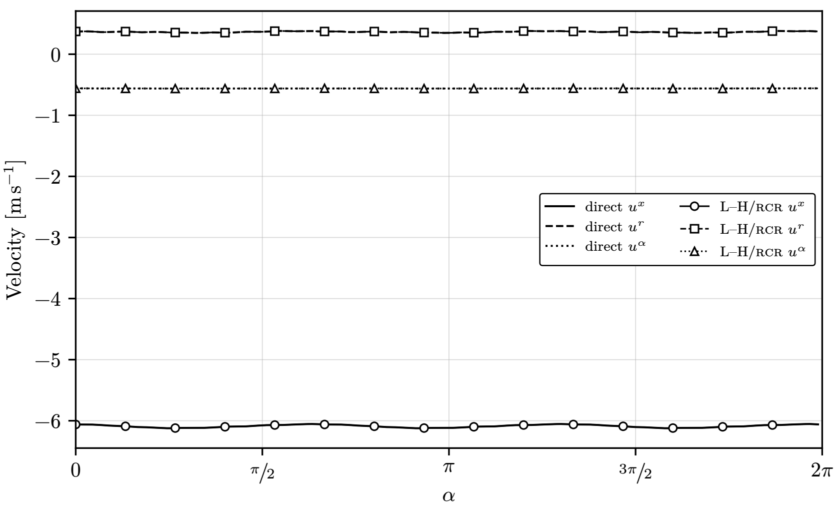


Figure 19: The induced velocity at ${}^{x}/_{D} = 1$, for ${}^{r}/_{R} = 0.8$.

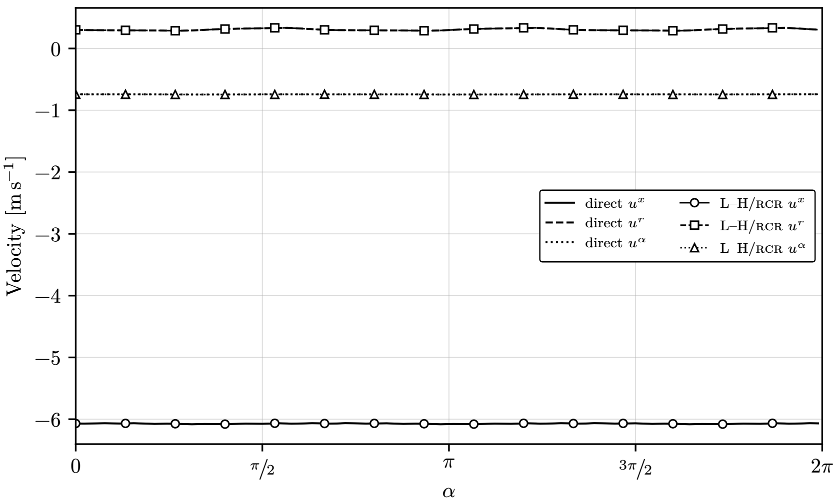


Figure 20: The induced velocity at $x/D = 1$, averaged radially.

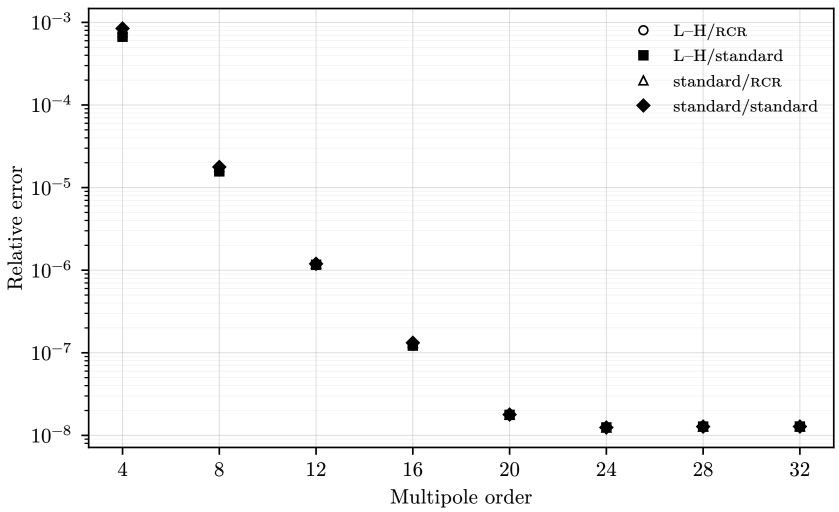


Figure 21: The relative error of the multipole induced velocity to the direct calculation at $x = 1D$.

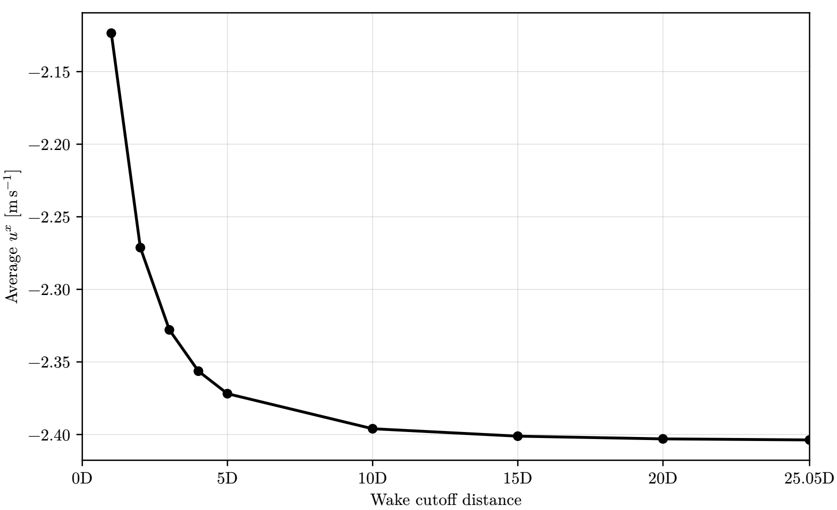


Figure 22: The velocity in the rotorplane along x calculated at different cutoff distances.

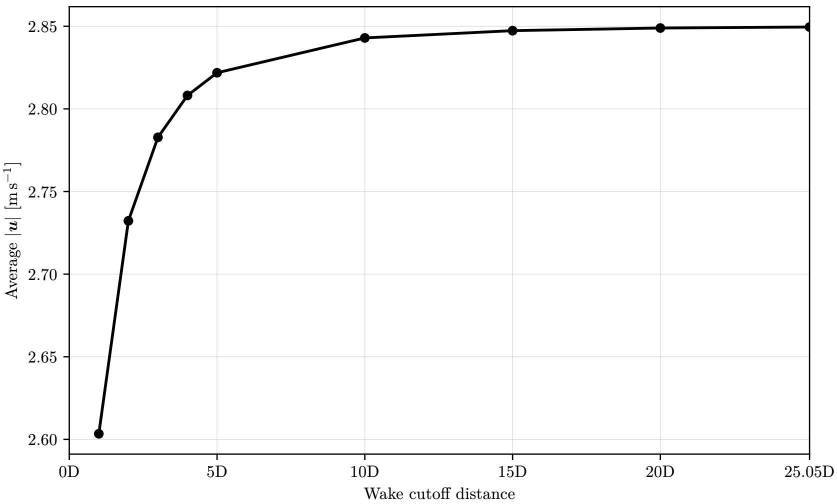


Figure 23: The weighted average of the velocity in the rotorplane calculated at different cutoff distances.

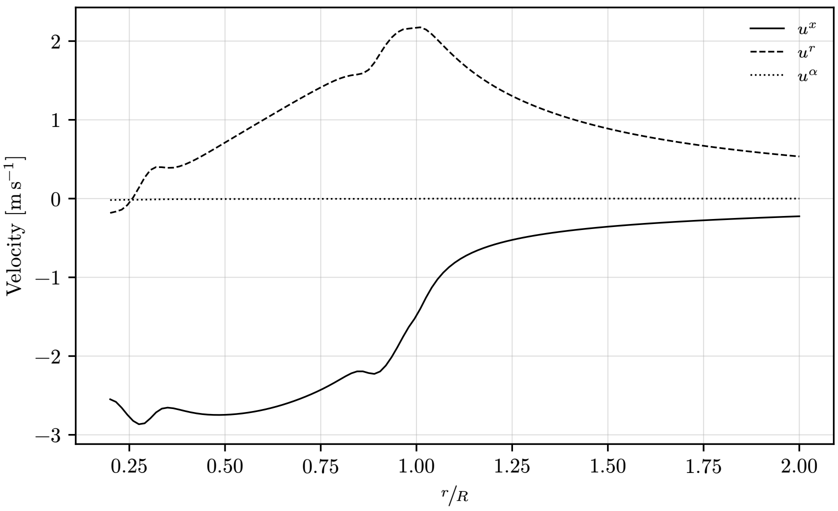


Figure 24: The average over the azimuth of all three cylindrical velocity components.

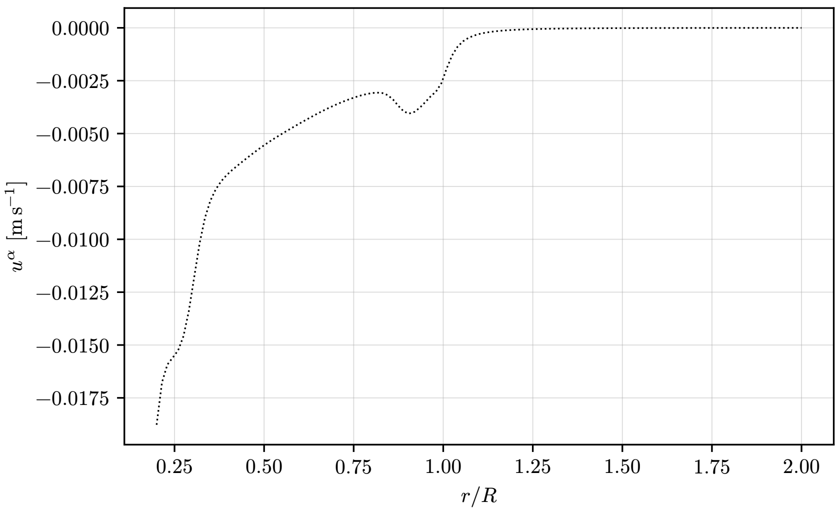


Figure 25: The average over the azimuth of the azimuthal velocity component.

7 Conclusion

The preceding mathematical discussion has revealed holes in the canopy of the methods considered through which beams of noumenal amendments glaringly gleam through. Most obvious is perhaps in the evaluation of the line integrals one must calculate, two nodes accounting for two distinct vortex filaments, one in each orientation. We showed that the mathematical nature of the vortex lattice method was this very notion of differences in potential jump across panels results in a circulation on the shared edge. For in truth, the circulation we do use to calculate the induced velocity is only the proverbial half of the circulation when not accounting for the inverse orientation. Accounting for this will effectively improve the calculation time by a factor of two.

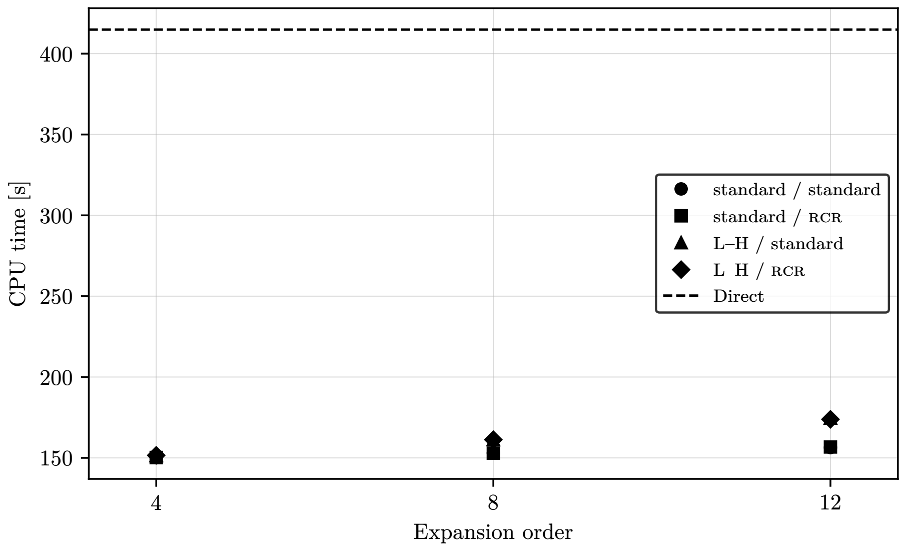


Figure 26: Median times of three benchmark CPU time measurements, comparing all fast multipole methods to the direct calculation.

As for the choices made during the performed tests, there is a multiplicity of factors to account for. Perhaps the most direct, yet elusive such factor is the assessment in determining the profundity and capacities of the octree partitioning. Increasing the profundity will of course decrease the number of direct calculations one must make. Yet, the number of translations one must perform is counteractively raised. Increasing the capacity will have the opposite effect, for if there are more permitted filaments in a given voxel, there will be more direct evaluations.[1] There is also the matter of choosing the velocity formulation and the multipole trnaslator. Both wall time measurments and measurements of the central processing unit show that the fast multipole method indeed is exceedingly faster that the direct calculation of the velocity, as is illustrated in figure 26, and tabulated in table 1.

One avenue that admittedly was the original intention of this thesis was that of dipole panel evaluation, as described in a paper of Deng *et alii*. This work does not necessarily employ the vortex lattice method, as it retains the dipole formulation of the integral equation, exploiting fast algorithms for solving the integrals upon the panel. From the analysis of the methodology, it seems the choice of a uniquely determined normal vector upon a plane panel. Given the fact that the unsteady vortex lattice method convects the nodes of the filaments that make up the vortex lattices, we have no guarantee that these panels are planar. In fact, most panels are *not* planar.[2] Unfortunately, grasping the mathematical admissibility of this issue could not be addressed in due time.

[1] [34] Gumerov & Duraiswami (2006)

[2] [47] Kebbie-Anthony *et al.* (2019)

Method	Order	Wall time [s]	CPU time [s]
direct	–	415.547	414.861
standard / standard	4	150.498	150.247
standard / standard	8	153.272	153.025
standard / standard	12	156.675	156.413
standard / RCR	4	150.480	150.230
standard / RCR	8	153.145	152.890
standard / RCR	12	157.005	156.741
L–H / standard	4	151.684	151.430
L–H / standard	8	161.601	161.294
L–H / standard	12	174.946	174.582
L–H / RCR	4	151.838	151.580
L–H / RCR	8	161.576	161.270
L–H / RCR	12	174.183	173.818

Table 1: Rotorplane benchmark timing.

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The Cyrillic Alphabet

az	а	А	*on*	о	О
buky	б	Б	*pokoi*	п	П
vede	в	В	*rci*	р	Р
glagoli	г	Г	*slovo*	с	С
ġe	ғ	Ғ	*tvrdo*	т	Т
dobro	д	Д	*će*	ћ	Ћ
đerv	ђ	Ђ	*uk*	у	У
je	е	Е	*frt*	ф	Ф
živete	ж	Ж	*her*	х	Х
zemlja	з	З	*ci*	ц	Ц
ež	ȝ	Ȝ	*črv*	ч	Ч
iže	и	И	*dželo*	џ	Џ
jot	й	Й	*ša*	ш	Ш
kako	к	К	*jery*	ы	Ы
l'udie	л	Л	*jeri*	ь	Ь
lje	љ	Љ	*èst*	э	Э
myslite	м	М	*ju*	ю	Ю
naš	н	Н	*ja*	я	Я
nje	њ	Њ	*ižica*	ѵ	Ѵ

The Greek Alphabet

alfa	α	A	*nu*	ν	N
beta	β	B	*xi*	ξ	Ξ
gamma	γ	Γ	*omikron*	o	O
delta	δ	Δ	*pi*	π	Π
epsilon	ε	E	*ro*	ρ	P
zeta	ζ	Z	*sigma*	σ ς	Σ
eta	η	H	*tau*	τ	T
theta	θ	Θ	*upsilon*	υ	Υ
iota	ι	I	*fi*	φ	Φ
kappa	$\varkappa$	K	*chi*	χ	X
lambda	λ	Λ	*psi*	ψ	Ψ
mu	μ	M	*omega*	ω	Ω

8 Appendix

8.1 Medial axes

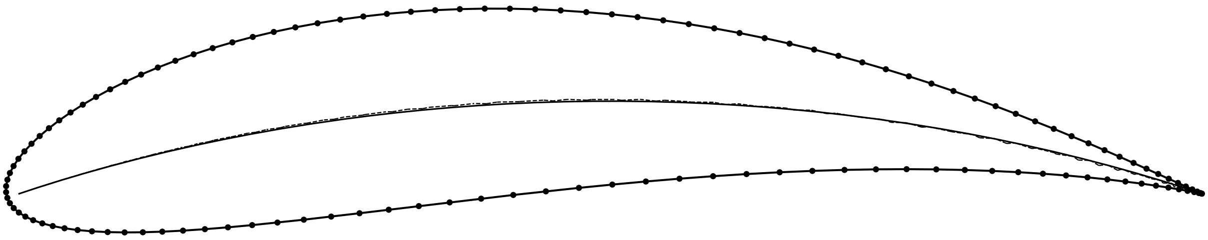

Figure 27: Medial axis of the von Kármán–Trefftz airfoil

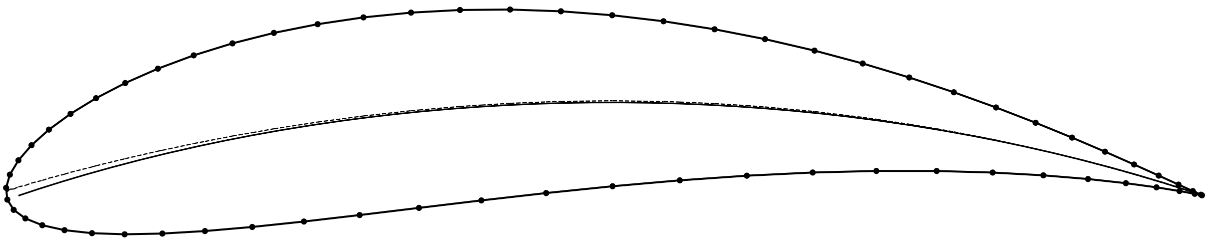

Figure 28: Mean camber line with skeleton of the von Kármán–Trefftz airfoil

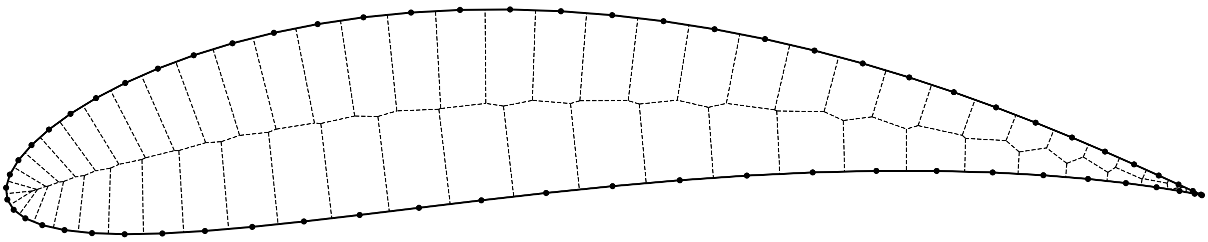

Figure 29: Voronoj diagram of the von Kármán–Trefftz airfoil

8.2 Algorithms

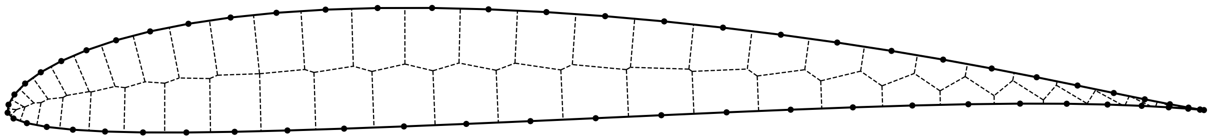

Figure 30: VORONOJ diagram of SD7032

Figure 31: Medial axis of SD7032

Algorithm 1 Octree construction

Require: Filaments $\{\boldsymbol{x}^{\mathrm{I}}_j, \boldsymbol{x}^{\mathrm{II}}_j, \Gamma_j\}_{j=1}^{\mathtt{N_s}}$, targets $\{\boldsymbol{ж}_i\}_{i=1}^{\mathtt{N_t}}$, capacity $\mathtt{n}$, maximal profundity $\mathtt{d}_{\mathrm{max}}$
Ensure: An octree whose voxels own source indices, target indices, and parent-child links
  $\boldsymbol{\ae}_j \leftarrow {}^1\!/\!_2(\boldsymbol{x}^{\mathrm{I}}_j + \boldsymbol{x}^{\mathrm{II}}_j)$ for each filament j
  Build the root voxel from all endpoints and all targets
  The root owns all source indices j and all target indices i
  SUBDIVIDE(root)
  function SUBDIVIDE($\mathtt{v}$)
    if $|\mathtt{sources}(\mathtt{v})| \leq \mathtt{n}$ or $\mathtt{depth}(\mathtt{v}) = \mathtt{d}_{\mathrm{max}}$ **then**
      Mark $\mathtt{v}$ as a leaf and **return**
    end if
    Create the eight child voxels of $\mathtt{v}$
    Initialize empty child source-index lists and child target-index lists
    for all $j \in \mathtt{sources}(\mathtt{v})$ **do**
      Compare both endpoints of filament j with the center of voxel $\mathtt{v}$
      if the endpoint extent lies in one child voxel octant **then**
        Assign j to that child voxel
      else
        Keep j owned by $\mathtt{v}$
      end if
    end for
    for all $i \in \mathtt{targets}(\mathtt{v})$ **do**
      Assign i to the child voxel octant containing $\boldsymbol{ж}_i$
    end for
    Replace $\mathtt{sources}(\mathtt{v})$ by the parent-owned filament list
    for all nonempty child voxel $\mathtt{c}$ **do**
      Set $\mathtt{parent}(\mathtt{v}) \leftarrow \mathtt{v}$ and recurse with SUBDIVIDE($\mathtt{c}$)
    end for
    if $\mathtt{v}$ has no nonempty child **then**
      Mark $\mathtt{v}$ as a leaf
    end if
  end function

Algorithm 2 Adjacency and adaptive interaction lists

Require: Octree $\mathtt{T}$, reach parameter $\mathtt{R} > 0$, target leaves
Ensure: Neighbor lists and adaptive $\mathtt{U}$, $\mathtt{V}$, $\mathtt{W}$, $\mathtt{X}$ source-voxel lists

```
function Adjacent(b, c)
    return |o_b − o_c| ≤ R√3 (r_b + r_c)
end function
a ← all target leaves and all their ancestors
for all b ∈ a do
    neighbors(b) ← {c : depth(c) = depth(b) and Adjacent(b, c)}
end for
for d = 1 to the actual profundity of T do
    for all b ∈ a with depth(b) = d do
        Clear V(b) and X(b)
        for all p ∈ neighbors(parent(b)) do
            if p is a leaf then
                if not Adjacent(p, b) then
                    Add p to X(b)
                end if
            else
                for all child c of p at depth d do
                    if c ∉ neighbors(b) then
                        Add c to V(b)
                    end if
                end for
            end if
        end for
    end for
end for
for all target leaf b do
    U(b) ← {c : c is a source leaf and Adjacent(b, c)}
    Clear W(b)
    for all c ∈ neighbors(b) with c ≠ b and c not a leaf do
        GatherW(b, c)
    end for
end for
function GatherW(b, c)
    if c is a leaf then
        if not Adjacent(b, c) then
            Add c to W(b)
        end if
        return
    end if
    for all child e of c do
        if not Adjacent(b, e) then
            Add e to W(b)
        else
            GatherW(b, e)
        end if
    end for
end function
```

Algorithm 3 Upward pass with M2M

Require: Octree `T`, filament data, truncation `N`, formulation map `SourceMap`
Ensure: Multipole expansion `multipole(v)` for every source voxel `v`

  for `d` = actual profundity of `T` down to 0 **do**
    for all voxel `v` at depth `d` **do**
      if `sources(v)` is not empty **then**
        Build `native(v)` from the filaments owned by `v`, centered at $\boldsymbol{o}_{\mathtt{v}}$
        Use Q2X followed by `SourceMap` for the chosen formulation
        $\mathtt{multipole(v)} \leftarrow \mathtt{multipole(v)} + \mathtt{native(v)}$
      end if
      if `v` has a parent and `multipole(v)` exists **then**
        $\mathtt{p} \leftarrow \mathtt{parent(v)}$
        $\mathtt{M} \leftarrow \mathrm{M2M}(\mathtt{multipole(v)}, \boldsymbol{o}_{\mathtt{p}} - \boldsymbol{o}_{\mathtt{v}})$
        Apply the post-M2M correction to `M` for Lamb–Helmholtz
        $\mathtt{multipole(p)} \leftarrow \mathtt{multipole(p)} + \mathtt{M}$
      end if
    end for
  end for

Algorithm 4 Downward pass with M2L and L2L

Require: Octree $\mathtt{T}$ after algorithm 3, target interaction lists
Ensure: Local expansions on target leaves

```
if some internal voxel owns native filaments then
    for all target leaf v do
        Initialize near(v) ← ∅ and a stack containing the root
        while the stack is not empty do
            Pop a source voxel c
            if c has no multipole and owns no sources then
                Continue
            end if
            if ADJACENT(v, c) then
                Add sources(c) to near(v)
                Push the children of c onto the stack
            else
                local(v) ← local(v) + M2L(multipole(c), o_v − o_c)
                Apply the post-M2L correction the translated local expansion
            end if
        end while
    end for
    return
end if
Build active target voxel interactions by algorithm 2
for d = 1 to the actual profundity of T do
    for all active target voxel v at depth d do
        if local(parent(v)) exists then
            local(v) ← local(v) + L2L(local(parent(v)), o_v − o_parent(v))
            Apply the post-L2L correction to the translated local expansion
        end if
        for all c ∈ V(v) ∪ X(v) do
            local(v) ← local(v) + M2L(multipole(c), o_v − o_c)
            Apply the post-M2L correction to the translated local expansion
        end for
    end for
end for
for all target leaf v do
    for all c ∈ W(v) do
        local(v) ← local(v) + M2L(multipole(c), o_v − o_c)
        Apply the post-M2L correction to the translated local expansion
    end for
end for
```